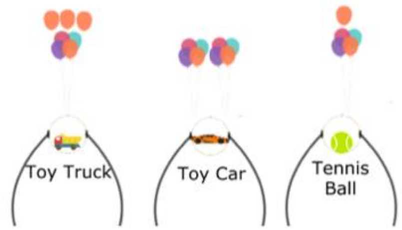
Unit 5 - Ratios and Rates		
Lesson	**Teacher Edition Page Number**	**Student Edition Page Number**
Unit 5 Overview	318	n/a
5.1 What Are Ratios?	325	173
5.2 Using Models to Represent Ratios	336	178
5.3 Exploring Rates	346	183
5.4 Exploring Tables of Equivalent Ratios	355	188
5.5 Connecting Ratios and Rates	365	193
5.6 Using Two- and Three-Column Tables of Equivalent Ratios	375	198
5.7 Digging Deeper Into Unit Rates	387	203
5.8 Applying Ratio Relationships to Percentages	398	209
5.9 Applying Ratios to Solve Problems Involving Percentages	408	214
5.10 Converting Units Using Ratios	417	219
5.11 Solving Problems Involving Ratios	428	224

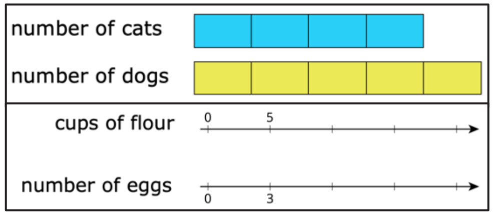
Unit 5 Overview: Ratios and Rates

Unit Narrative

Students engage with ratios in a variety of contexts and apply an understanding of part-to-part and part-to-whole relationships for everything from mixtures and recipes (Lesson 2) to making objects of different weights float with balloons (Lesson 4) to solving problems with percentages (Lessons 8-9) to unit conversions within systems (Lesson 10). Mathematical Thinking and Reasoning Standard 2 (MA.K12.MTR.2.1) is a focus throughout this unit, with students using a variety of visual models and representations, including double number lines, tape diagrams, and two-column and three-column ratio tables.



In previous grades, students worked with fractions as a way to represent the division of two quantities. Students extend this interpretation of a fraction to represent a part-to-whole relationship between two quantities as a ratio while also learning that ratios can be used to represent part-to-part relationships as well. Students learn different ways of representing a ratio, including $a: b$, a to b, or $\frac{a}{b}$.



The unit introduces ratios using a variety of visual models and with activities that explore and compare quantities in real-world contexts. Students learn and explore a variety of representations in this unit to develop a strong foundational understanding of these concepts and to increase entry points for all students. Their knowledge of ratios sets the foundation that extends to solving multi-step problems with proportional relationships in Unit 12 and linear relationships in Grade 7 Accelerated Mathematics.

Instruction includes distinguishing between part-to-part and part-to-whole ratios and understanding the difference between ratios and fractions. Students are introduced to percentages in Unit 2 of Grade 6 Accelerated. In this unit, they will extend that understanding to connect percentages as part-to-whole ratios. Students will use this understanding to solve problems involving percentages later in the course in Unit 12.

Unit At-A-Glance

Section 1: Introduction to Ratios	Benchmarks Addressed	Lesson 1	What Are Ratios?
	MA.6.AR.3.1	Lesson 2	Using Models to Represent Ratios
Section 2: Connecting Rates to Ratios	Benchmarks Addressed	Lesson 3	Exploring Rates
	MA.6.AR.3.2 **MA.6.AR.3.3**	Lesson 4	Exploring Tables of Equivalent Ratios
		Lesson 5	Connecting Ratios and Rates
Section 3: Using Tables of Equivalent Ratios	Benchmarks Addressed	Lesson 6	Using Two- and Three-Column Tables of Equivalent Ratios
	MA.6.AR.3.2 **MA.6.AR.3.3** **MA.6.AR.3.5**	Lesson 7	Digging Deeper Into Unit Rates
Section 4: Percentages and Other Real-World Contexts	Benchmarks Addressed	Lesson 8	Applying Ratio Relationships to Percentages
	MA.6.AR.3.4 **MA.6.AR.3.5**	Lesson 9	Applying Ratios to Solve Problems Involving Percentages
		Lesson 10	Converting Units Using Ratios
		Lesson 11	Solving Problems Involving Ratios

Differentiation

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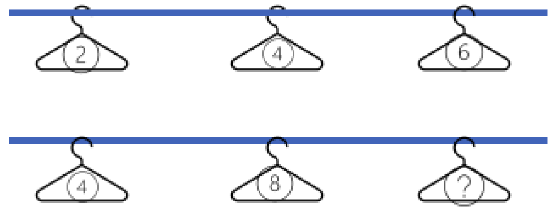
Engagement The "WHY" of learning	• Token Economy (Ticket Booth): The Warm-Up in Lesson 7 highlights a common real-world application of ratios—tickets at a carnival or fair. Consider implementing a true token economy in your classroom. Students can earn tickets for exemplifying the soft skills of Engagement in UDL; for example, students who show growth from one lesson to another or students who showcase self-reflection like in Lesson 7, 5.7.5 Bring It Together. Not only will this practice promote, recognize, and praise students' actions like sustained effort, persistence, and self-regulation, but also students actually having to convert their number of tickets to prizes they can earn shows in-the-moment application of rates and ratios as they learn them in the unit.
Representation The "WHAT" of learning	• Try a clothesline double number line in the classroom for a creative and engaging representation of double number lines as ratios. Continuously offer unique and creative ways for students to customize the display of their information. In this unit, that means students opting into the visual model that makes the most sense to them, whether that be a double number line, a tape diagram, equivalent ratio tables, or something else. 
Action & Expression The "HOW" of learning	• Consider an extension project for additional opportunities to engage students with different ways to demonstrate what they know about rates and ratios. Lesson 6 highlights mathematician Mary Jackson's contributions to NASA while calculating by hand before the introduction of the IBM machine. Consider project-based learning for students to explore engineering contributions like Mary Jackson's and how they may have used rates and ratios in their "human computer" calculations. • My Favorite Recipe: Utilize "Nana's Chocolate Milk" Warm-Up in Lesson 2 as a launch for project-based home learning. Students can create and produce their own recipes at home to demonstrate their understanding of ratios in real-world contexts. Consider having students use a variety of tools like pictures or video apps to express their recipes.

ELL Information

Sentence Frames Sentence frames offer guided scaffolding for English Language Learners developing their skills in language acquisition and math literacy. Students consistently engage in sensemaking with ratios, rates, and proportions throughout the unit, differentiating between what each term means and their application in context. Provide sentence frames for English Language Learners as scaffolding for constructing sentences and concepts related to the following:

- Lesson 1: What Are Ratios?
- Lesson 5: Connecting Ratios and Rates
- Lesson 7: Digging Depper Into Unit Rates
- Lesson 8: Applying Ratio Relationships to Percentages
- Lesson 10: Converting Units Using Ratios

This ________
Rate
Ratio
Percentage

means I have _______ _______

for every _______ _______.

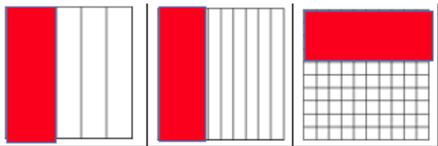
1 | 3 | 9 | 4 | dollars | centimeters | rate | yard | inches | hour

Additional Differentiated Support

Scaffolding Support

Mathematical Thinking and Reasoning Standard 2 (MA.K12.MTR.2.1): Multiple Representations
Double number lines, tape diagrams, and equivalent ratio tables provide students with three unqiue options for visually expressing different ratios.
Notice Benchmark **MA.6.AR.3.3** is the only benchmark that cites a specific type of representation (table), so continue to spiral opportunities for students to choose their own representations to solve problems.

Statement	Model
20% of 60 is 12.	0, 12, 24, 36, 48, 60 0%, 20%, 40%, 60%, 80%, 100%
48% of 25 is 12.	Value: 25, 12 / Percentage: 100, 48
40% of 70 is 28.	value: 7, 7, 7, 7, 7, 7, 7, 7, 7, 7 %: 10, 10, 10, 10, 10, 10, 10, 10, 10, 10
16% of 125 is 20.	Value: 125, 20 / Percentage: 100, 16 *Answers will vary.*

Model			
Fraction	$\frac{2}{5}$	$\frac{4}{10}$	$\frac{40}{100}$
Percentage	$\frac{2}{5} \times 100 = 40\%$	$\frac{4}{10} \times 100 = 40\%$	$\frac{40}{100} \times 100 = 40\%$

For example, notice the synthesis in Lesson 8, the 5.8.5 Your Turn! activity, as it positions students as experts in all three representations as a way to synthesize their learning so far in the unit.

Consider providing students with manipulatives, coordinate planes with gridlines to use as boxes, or colored pencils for a variety of tools to represent different rates or ratios.

On-Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Graph Two-Variable Relationships
- Create a Table for Two-Variable Relationships
- Mixed Numbers and Unit Fractions
- Generate a Number Pattern
- Number Lines: Whole Number Sums and Differences
- Number Lines: Mixed Numbers
- Number Lines: Fractions
- Convert from a Larger to Smaller Unit
- Convert from a Smaller to Larger Unit
- A Fraction is the Quantity Formed by Its Parts
- Understanding a Unit Fraction
- Division of a Whole Number by a Unit Fraction
- Division of a Whole Number by a Unit Fraction in Real-World Problems
- Recognize and Generate Equivalent Fractions Using Models
- Representing Fractions on a Number Line
- Multiply a Fraction by a Whole Number

Section Overview

Section 1: Introduction to Ratios (Lessons 1-2)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="2">MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a\!:b$, where $b \neq 0$.
• Clarification 1: Instruction focuses on the understanding that a ratio can be described as a comparison of two quantities in either the same or different units.
• Clarification 2: Instruction includes using manipulatives, drawings, models and words to interpret part-to-part ratios and part-to-whole ratios.
• Clarification 3: The values of a and b are limited to whole numbers.</td><td>Lesson 1</td><td>• I can write and interpret ratios.
• I can draw a diagram that represents a ratio and explain what a diagram means.</td></tr>
<tr><td>Lesson 2</td><td>• I can write and interpret ratios.
• I can represent ratios using various models.</td></tr>
</table>

Section 2: Connecting Rates to Ratios (Lessons 3-5)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="3">MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.
• Clarification 1: Instruction includes using manipulatives, drawings, models and words and making connections between ratios, rates and unit rates.
• Clarification 2: Problems will not include conversions between customary and metric systems.

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.
• Clarification 1: Instruction includes using two-column tables (e.g., a relationship between two variables) and three-column tables (e.g., part-to-part-to-whole relationships) to generate conversion charts and mixture charts.</td><td>Lesson 3</td><td>• I can use ratios and rates to describe relationships between quantities.
• I can write a ratio as a unit rate.</td></tr>
<tr><td>Lesson 4</td><td>• I can build tables of equivalent ratios incorporating an understanding of unit rate.</td></tr>
<tr><td>Lesson 5</td><td>• I can make connections between ratios, rates, and unit rates.</td></tr>
</table>

Section 3: Using Tables of Equivalent Ratios (Lessons 6-7)

Benchmarks Addressed	Learning Targets	
MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate. • *Clarification 1: Instruction includes using manipulatives, drawings, models and words and making connections between ratios, rates and unit rates.* • *Clarification 2: Problems will not include conversions between customary and metric systems.* **MA.6.AR.3.3** Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios. • *Clarification 1: Instruction includes using two-column tables (e.g., a relationship between two variables) and three-column tables (e.g., part-to-part-to-whole relationships) to generate conversion charts and mixture charts.* **MA.6.AR.3.5** Solve mathematical and real-world problems involving ratios, rates and unit rates including comparisons, mixtures, ratios of lengths and conversions within the same measurement system. • *Clarification 1: Instruction includes the use of tables, tape diagrams and number lines.*	Lesson 6	• I can use two- and three-column tables to display equivalent ratios. • I can solve problems using two- and three-column tables of equivalent ratios.
	Lesson 7	• I can determine a unit rate when given different ratios. • I can interpret unit rates in real-world contexts. • I can use unit rates to solve problems.

Section 4: Percentages and Other Real-World Contexts (Lessons 8-11)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="4">MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.
• Clarification 1: Instruction includes the comparison of $\frac{part}{whole}$ to $\frac{percent}{100}$ in order to determine the percent, the part or the whole.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.
• Clarification 1: Instruction includes the use of tables, tape diagrams and number lines.</td><td>Lesson 8</td><td>• I can make connections between ratios and percentages.
• I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.</td></tr>
<tr><td>Lesson 9</td><td>• I can apply ratio relationships to solve problems involving percentages.</td></tr>
<tr><td>Lesson 10</td><td>• I can use ratios to convert units within the same measurement system.</td></tr>
<tr><td>Lesson 11</td><td>• I can solve problems involving ratios, rates, and unit rates.</td></tr>
</table>

Vertical Alignment

	Grade 6 Accelerated Unit 5 Rates and Ratios	Grade 7 Accelerated Unit 5 and Unit 7 Solving Multi-step Proportional Relationships Representing Proportional Relationships
Prior Course(s)	**Course Level**	**Future Course(s)**
MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction. **MA.5.AR.3.1** Given a numerical pattern, identify and write a rule that can describe the pattern as an expression. **MA.5.M.1.1** Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.	**MA.6.AR.3.1** Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a\!:\!b$, where $b \neq 0$. **MA.6.AR.3.2** Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate. **MA.6.AR.3.3** Extend previous understanding of fractions and numerical patterns to generate or complete a two- or thee-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios. **MA.6.AR.3.4** Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities. **MA.6.AR.3.5** Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.	**MA.7.AR.4.2** Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship. **MA.7.AR.4.4** Given any representation of a proportional relationship, translate the representation to a written description, table, or equation. **MA.7.AR.3.3** Solve mathematical and real-world problems involving the conversion of units across different measurement systems. **MA.7.AR.4.5** Solve real-world problems involving proportional relationships.

5.1 What Are Ratios?

Lesson Introduction

Benchmark	MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a: b$ where $b \neq 0$.	Learning Targets	• I can write and interpret ratios. • I can draw a diagram that represents a ratio and explain what the diagram means.

Benchmark Coherence	Building On	Working Toward
	MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.	N/A

Essential Questions	• What is a ratio? • What are three ways to write ratios? • Can a ratio be simplified? • Explain if the ratios are the same if you switch the numbers.
Lesson Narrative	In this first lesson on ratios, students are introduced to several different ways to describe a situation using ratio language. Initially, students sort figures by color, area, and a third attribute of their choice. Next, students learn how to write and interpret ratios and then practice these skills. In the 5.1.4 Guided Instruction, students learn to represent and interpret ratios with diagrams. Finally, students reflect on their ability to write, draw diagrams of, and interpret ratios.

Component Summary				
	5.1.1 Warm-Up (~5 min.)	Sort figures by color, area, and a third attribute of their choice *(SE p. 173)*	5.1.4 Guided Instruction (10 min.)	Represent ratios with diagrams *(SE p. 176)*
	5.1.2 Guided Instruction (15 min.)	Write and interpret ratios *(SE p. 174)*	5.1.5 Collaboration (10-15 min.)	Interpret ratios with diagrams *(SE p. 177)*
	5.1.3 Try It (5-10 min.)	Students practice writing and interpreting ratios *(SE p. 175)*	5.1.6 Wrap-Up (~5 min.)	Students reflect on their ability to write, draw diagrams of, and interpret ratios *(SE p. 178)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms</u>: • Ratio <u>Additional Terms:</u> N/A	• (Optional) Note cards • (Optional) Chart paper • (Optional) Markers	• Consider creating an anchor chart.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

Students may write ratios with no descriptive words or units. For example: $a{:}\,b$, and for some numbers, a and b.
Students may not pay close attention to the order of the quantities, but the order of the quantities impacts the ratio. For example, $a{:}\,b$ is not the same as $b{:}\,a$, for some numbers a and b.
Students may think ratios and fractions are the same when they are not. Fractions compare part-to-whole relationships while ratios can compare part-to-part or part-to-whole relationships.

</td><td>

In the Warm-Up, the teacher should differentiate between the small L-shaped figures and the large L-shaped figures.
Writing a ratio as $a{:}\,b$, for some numbers a and b is a good start, but part of writing a ratio is stating what those numbers mean. Draw students’ attention to the sentence stems in the task statement and encourage them to use those words.
Create a reference poster and allow students to create reference cards with instructions and examples to aid them in writing and interpreting ratios represented with diagrams.

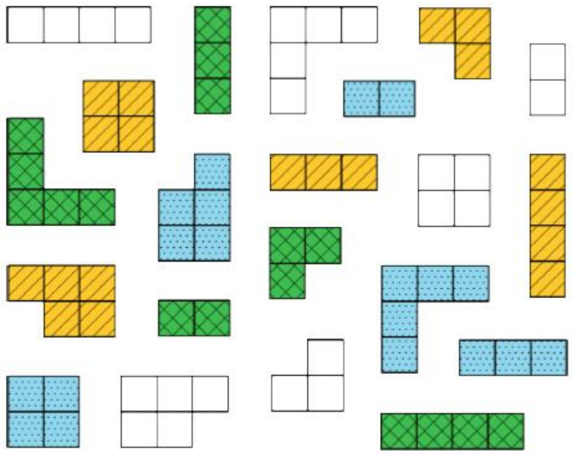
</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
The Think-Pair-Share routine can be used in components 5.1.1-5.1.4 of the lesson. Initially, have students attempt the questions individually. Next, pair students up and ask them to discuss their strategies, reasoning, and answers. Tell students to resolve any differences in their answers. This routine helps students build their confidence and understanding.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Throughout the lesson, students connect mathematical concepts to everyday experience. They model real-world scenarios with ratios. They also analyze and describe diagrams that represent real-world ratio relationships. Students must critically think about the order of the quantities and whether they are asked to find a part-to-part or a part-to-whole ratio.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider having students create a reference card with an explanation of how to write and interpret ratios represented in diagrams. Encourage students to include at least one example of each process. Allow students to use their reference cards during the lesson. The diagrams of the ratios provide a visual entry point. Structuring components with the Think-Pair-Share routine provides an auditory entry point.
Mark-Up Text: Notice the use of Mark-Up on a number line in the Wrap-Up. Consider utilizing Mark-Up during lesson components for students to notate if they are uncertain about a portion of the lesson or if they feel confident answering the questions.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.1.1 Warm-Up: What Kind and How Many?

Component Overview: *In this Warm-Up, students compare figures and sort them into categories. The first two questions are straightforward to allow all students access to the component and prime them to think about other ways the figures can be sorted. Students create their own categories in the third question and explain their reasoning.*

5.1.1 Warm-Up: What Kind and How Many?

A set of figures with different colors, shapes, and sizes is shown.



The figures can be sorted using different categories.

1. Complete the table by determining the number of groups for each method of sorting. For the last row, determine a third way the figures could be sorted and how many groups the sorting method would create.

Sorting Method	Number of Groups Created
Sorted by color	4
Sorted by area	4
Sorted by *Answers will vary* *Sorted by shape*	*Answers will vary* 7

Think-Pair-Share
This Warm-Up provides an opportunity to engage students in discourse to discover the multiple ways the figures can be sorted. Engage students in a Think-Pair-Share after completing the table. Have students explain how they sorted the figures. Emphasize that the important thing is to describe the way they sorted them clearly enough, so everyone agrees it is a reasonable way to sort them. Tell students we will be looking at different ways of seeing the same set of objects in the next component.

5.1.2 Guided Instruction: Writing and Interpreting Ratios

Component Overview: *In this exploration, students first complete statements to describe the relationships between the figures in the Warm-Up. Students learn that they can express the ratio of a to b three ways:* $\boldsymbol{a}:\boldsymbol{b}$, $\boldsymbol{a}$ *to* $\boldsymbol{b}$, *or* $\frac{a}{b}$. *Finally, students write and interpret ratios measuring the amount of chai powder used in recipes for chai tea.*

5.1.2 Guided Instruction: Writing and Interpreting Ratios

The figures from the Warm-Up can be described using ratio language.

A ***ratio*** is a comparison of two or more quantities.

1. Complete the statements to describe the relationships between the figures from the Warm-Up.
   a. There are 5 yellow figures for every 5 green figures.
   b. For every 3 figures with an area of 2 square units, there are 6 figures with an area of 5 square units.
   c. The ratio of square-shaped figures to L-shaped figures is 3 to 6.

Ratios can be written in three ways. The ratio of blue figures to white figures is expressed in all three ways below.

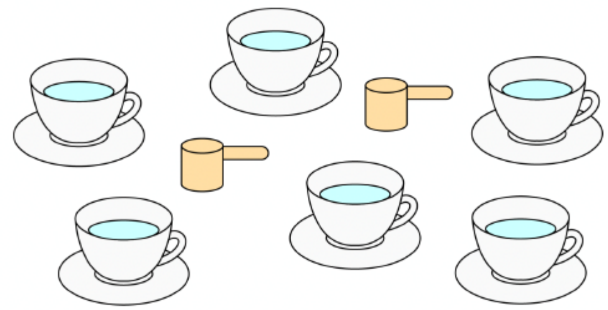
Method for Writing Ratios	The Ratio of Blue Figures to White Figures
Using a colon (:)	5 : 6
Using the word "to"	5 to 6
Using a fraction bar	$\frac{5}{6}$

This Guided Instruction is intentionally scaffolded, as this is the first-time students are working with ratios. Model your approach and language in questions 1 and 2 before giving students an opportunity to try it on their own in question 3.

Consider engaging students with a hook here to give them a creative opportunity to make sense of ratios — What are two quantities you would love to have in a ratio of $5:2$ *but hate to have in a ratio of* $2:5$*?*

5.1.2 Guided Instruction: Writing and Interpreting Ratios (continued)

2. Avril makes a tea called chai using a pre-made powdered mix. The box instructions say to mix 2 scoops of chai powder with 6 cups of hot water.



Complete each description of the collection by writing a number in each box.

a. The ratio of chai mix scoops to cups of water is

$\boxed{2} : \boxed{6}$.

b. The ratio of cups of water to chai mix scoops is

$\dfrac{\boxed{6}}{\boxed{2}}$.

Students may confuse writing the ratio of scoops to cups of water and cups of water to scoops. You may see students writing both ratios as $2: 6$ or $6: 2$, or students may mix up the numbers for each answer.
How many scoops of chai are needed to make the tea? What box should this go in if we are describing the ratio of scoops to cups of water?
How many cups of water are needed? What box should this go in if we are describing the ratio of cups of water to scoops?
Will the ratio of scoops to cups of water be the same as the ratio of cups of water to scoops? Why not?

5.1.2 Guided Instruction: Writing and Interpreting Ratios (continued)

3. The relationship between chai mix and water can also be described using other numbers. Consider the rearranged image.

Complete the statements.

a. There are 3 cups of water per scoop of chai mix.

b. There is $\frac{1}{3}$ scoop of chai mix for every cup of water.

Notice the fraction bar for question 3b. Notice how students articulate their responses, as this is not actually a fraction of a part:whole relationship but instead a ratio representing a part:part relationship.
Consider pausing to ask students whether something represents a part:part or part:whole relationship.

The order of the ratio is important, as it relates specifically to the quantities being described. The order used in the written description must match the order of the values that are being compared.

The built-in partner and small-group discussions provide an entry point for auditory learners. To engage visual learners, encourage students to label the ratio boxes with different colors, one for scoops and one for cups of water. This can help students keep track of how to write the ratio correctly.

4. Discuss with your partner why the ratio of chai mix scoops to cups of water is not equivalent to the ratio of cups of water to chai mix scoops.
Answers will vary. There are a different number of chai mix scoops and cups of water. The order matters when describing ratios. When referring to values like 1 and 3, I have to keep them in order so I know which value goes with which unit.

Consider providing sentence stems for additional scaffolding for question 4.
The ratios are not equivalent because there are a different number of ______ and ______.
The order (does/does not) matter when writing ratios.
When I write ratios, I need to pay attention to _____.

5.1.3 Try It: Writing and Interpreting Ratios

Component Overview: *Students work with a partner to write and interpret ratios in a given context.*

5.1.3 Try It: Writing and Interpreting Ratios

In a fruit bowl, there are 6 oranges, 4 apples, and 2 plums.

1. Complete the statements to describe the contents of the fruit bowl.

   a. The ratio of oranges to apples is 6 : 4.

   b. The ratio of plums to apples is 2 to 4.

   c. For every 4 apples, there are 2 plums.

   d. For every 3 oranges, there is one
      - ○ apple.
      - ○ plum.

2. Explain how you determined the answer for Part d using words or a sketch.

   Divide the 6 oranges into equal groups of 3. There are 2 groups of 3 oranges. Then, divide the 4 apples into 2 groups, which resulted in 2 apples per group. Next, divide the 2 plums into 2 groups, which resulted in 1 plum per group.

6 Oranges		
4 Apples		
2 Plums		

During this component, students should be engaging in informal discussions to find the ratios of each type of fruit in the bowl. Circulate the room to listen for how students are determining each ratio and continue to use prompting questions to clarify misconceptions as they arise.

- *Where should you write the number of oranges if it is asking for the ratio of oranges to apples?*
- *How can you determine the ratio of 3 oranges to apples or plums? Is there a way to represent this visually?*
- *How can you show these ratios using a colon or fraction bar?*

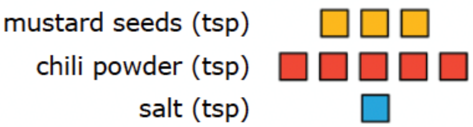
5.1.4 Guided Instruction: Representing Ratios With Diagrams

Component Overview: *Guide students through this component as they consider how to use ratios to represent part:part and part:whole relationships. Students work with a partner to represent the relationships between different ingredients in a recipe using ratios to represent the part to part and part to whole relationships.*

5.1.4 Guided Instruction: Representing Ratios with Diagrams

When solving problems involving ratios, it can be helpful to use diagrams and number lines to represent the relationships described.

1. A recipe for 1 batch of spice mix says, "Combine 3 teaspoons (tsp) of mustard seeds, 5 teaspoons of chili powder, and 1 teaspoon of salt." Below is a diagram to represent this situation.



   How many teaspoons of spices are in 1 batch of the mix?
   9 teaspoons

Ratios can be used to describe part-to-part relationships, where one part is being compared to another, and part-to-whole relationships, where one part is being compared to the entire group.

2. Complete the statements in the table.

Type of Relationship	Example Statements
part-to-part	The ratio of tsp of chili powder to tsp of mustard seeds is 5:3. There are 3 teaspoons of mustard seeds per tsp of salt.
part-to-whole	The ratio of teaspoons of chili powder to teaspoons of spice mix is 5 to 9. For every 1 teaspoon(s) of salt, 9 teaspoon(s) of spice mix can be made.

3. Discuss with your partner what is meant by "whole" in part-to-whole.

Consider having students create a reference card with an explanation of how to write and interpret ratios represented in diagrams. Encourage students to include at least one example of each process. Allow students to use their reference cards as a personal anchor chart during the lesson.

Compare and Connect
Consider using the Compare and Connect routine when students discuss their answers with their partners.

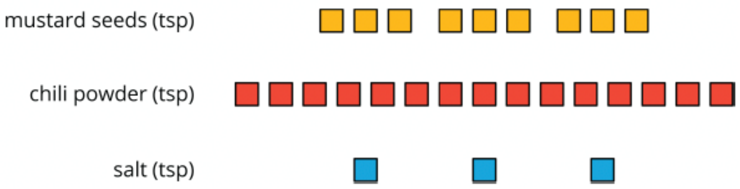
- *What is the same?*
- *What is different?*
- *What is meant by "whole" (question 3) and what does that make you think about writing ratios?*

5.1.5 Collaboration: Interpreting Ratios With Diagrams

Component Overview: *Students collaboratively practice writing and interpreting ratios of a spice mix recipe and a paint mixture that are represented with diagrams. The diagrams of the ratios provide a visual entry point. Structuring components with the Think-Pair-Share routine provides an auditory entry point.*

5.1.5 Collaboration: Interpreting Ratios with Diagrams

1. The diagram shows the ingredients Luca mixed to make the same spice mix recipe.



Work with your partner to determine how many batches of spice mix are represented by the diagram. Explain or show your reasoning.
Luca made 3 batches of spice mix. In Luca's mix, there are 9 teaspoons of mustard seeds. Each batch is to contain only 3 teaspoons of mustard seeds. 9 teaspoons of mustard seeds in Luca's mix $\div 3$ teaspoons in the recipe $= 3$ batches of spice mix.

	Mustard seeds	**Chili Powder**	**Salt**
tsp in Luca's mix	9	15	3
Recipe mix	3	5	1
Luca's mix ÷ Recipe mix	$9 \div 3$	$15 \div 5$	$3 \div 1$
Number of Batches of Spice Mix	$= 3$	$= 3$	$= 3$

The table is scaffolded in a way that guides students toward understanding the number of batches of spice mix represented by the diagram. To further guide students toward understanding, consider asking the following questions: How can you determine the number of batches? How does the table help us find the number of batches of spice mix? Is there a way to write a ratio to show this?

5.1.5 Collaboration: Interpreting Ratios With Diagrams (continued)

2. When mixing paint colors, the same shade is produced when the same ratios of each color are used to mix the paint. The diagram represents 4 batches of light green paint that Mabelise mixed.

white paint (cups)	□□□□□□□□□□□□
green paint (cups)	■■■■■■■■■■■■■■■■

Use the diagram to answer the following questions with your partner.

a. Draw a diagram that represents 1 batch of the same shade of light green paint.
Divide the number of cups of each paint color by 4.

white paint (cups)	□□□	□□□	□□□	□□□
green paint (cups)	■■■■	■■■■	■■■■	■■■■

b. Complete the statements to describe the relationships.

- For every 3 cups of white paint, there are 4 cups of green paint in the mix.
- The ratio of (○ white paint ○ green paint) to (○ white paint ○ green paint) is $\frac{4}{3}$.
- The ratio of white paint to the light green mixture is 3 to 7.
- There are 8 cups of green paint used for every 14 cups of light green paint that is mixed.

This component provides visual and auditory entry points for students through the use of diagrams and partner discussion. Encourage students to use the visual scaffolds to determine the different ratios in the table and connect them to the abstract representations of ratios using colons and fraction bars.

Partner pairs should make connections between the visual representation of the paint and writing this as a ratio. Students should be able to articulate the ratio of green to white paint through connection to the visual but may struggle to determine the ratio of white to light green. Encourage students to circle each batch of light green in the table to provide a visual representation of light green paint.

5.1.6 Wrap-Up: Reflection

Component Overview: *Students reflect on their understanding and confidence in writing and interpreting ratios, as well as drawing and explaining diagrams that represent ratios.*

5.1.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can write and interpret ratios.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary. Students may mark that they are "getting there but need more practice."

2. I can draw a diagram that represents a ratio and explain what a diagram means.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary. Students may mark that they are "getting there but need more practice."

Consider having students provide justification or evidence, either through written expression or visual demonstration.

5.2 Using Models to Represent Ratios

Lesson Introduction

Benchmark	MA.6.AR.3.1 Given a real-world context, write and interpret ratios to show the relative sizes of two quantities using appropriate notation: $\frac{a}{b}$, a to b, or $a : b$ where $b \neq 0$.	Learning Targets	• I can write and interpret ratios. • I can represent ratios using various models.

Benchmark Coherence	Building On	Working Toward
	MA.5.FR.1.1 Given a mathematical or real-world problem, represent the division of two whole numbers as a fraction.	N/A

Essential Questions	• How do you write ratios? • How do you interpret ratios? • How do you represent ratios using models?
Lesson Narrative	In this lesson, students begin by describing and comparing ratios of cups of milk to scoops of chocolate powder. They explore representing a ratio with a double number line and use it to interpret the ratio. Students make the connection between representing ratios using double number lines and tape diagrams, interpreting ratios through these visual representations.

Component Summary				
	5.2.1 Warm-Up (~5 min.)	Describe and compare ratios of cups of milk to scoops of chocolate powder *(SE p. 179)*	5.2.4 Guided Instruction (10 min.)	Students use tape diagrams to represent ratios *(SE p. 181)*
	5.2.2 Exploration (15 min.)	Explore representing a ratio with a double number line and use it to interpret the ratio *(SE p. 179)*	5.2.5 Your Turn! (10 min.)	Practice using double number lines and tape diagrams to represent and interpret ratios *(SE p. 182)*
	5.2.3 Try It (5-10 min.)	Practice using double number lines to represent and interpret ratios *(SE p. 180)*	5.2.6 Wrap-Up (~5 min.)	Students use a tape diagram to create a double number line and interpret a ratio *(SE p. 183)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Ratio	• Clothesline Number Lines	• Warm-Up 5.2.1 contains a video to accompany the question prompts.

Teacher Tips

Common Misconceptions

- Students may forget that ratios can be expressed three ways: a to b, a: b, or $\frac{a}{b}$.
- Students may forget to simplify ratios, especially when the ratio is not expressed using a fraction bar.
- Students may be confused when they see the same number in different positions in a double number line.

Things to Consider

- Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in writing ratios, interpreting ratios, and representing ratios using various models.
- Utilize the Warm-Up and Exploration as one cohesive, student-centered discovery of double number lines in real-world context.

Literacy and Language Routines

Notice and Wonder

5.2.1 Warm-Up and the 5.2.2 Exploration can be structured with the Notice and Wonder routine. Ask students what they notice and wonder about the difference between the ratio of chocolate powder to milk in the recipe and in Dan's mixture. Ask students to share and justify the reasoning behind their responses.

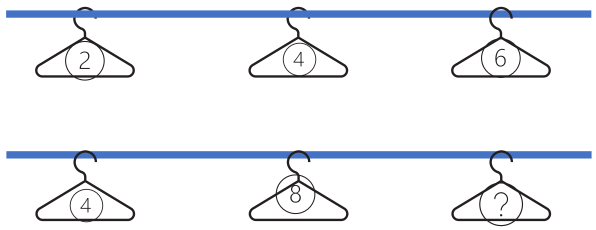
Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.2.1: Multiple Representations

Throughout the lesson students see ratios represented multiple ways. The ratios are expressed three ways: a to b, a: b, or $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout the lesson, deepening conceptual understanding.

Differentiate Instruction

Clothesline Number Lines offer manipulatives for students to interact with double number lines for hands-on learning. Consider using a kinesthetic representation of the double number lines on clotheslines that cross the room. Students can place and adjust numbers on folded paper/cardstock on the clothesline, either folding pieces of paper or using clothespins to hold each number in place.



Lesson Notes:

5.2.1 Warm-Up: Nana's Chocolate Milk

Component Overview: *Play the video for the class, then students work independently to describe and compare ratios of cups of milk to scoops of chocolate powder. Students can move on to 5.2.2 Exploration without getting a right answer to question 1b, as they will continue to work on this problem in the next component.*

5.2.1 Warm-Up: Nana's Chocolate Milk

1. Dan's Nana prefers her chocolate milk made with just the right "chocolateyness," but he made a mistake when mixing it for her. The image below shows what Dan should have mixed and what he actually mixed.

Students may initially respond with "Adding more milk" or "Adding more milk and more chocolate" – prompt them to use a ratio to find out how much more milk and/or chocolate.
Notice 5.2.2 Exploration continues this concept with a double number line.

   a. Describe what Dan did wrong.

   Dan added one extra scoop of chocolate.

   b. How could Dan fix it so that the glass contains Nana's preferred balance of milk and chocolate?

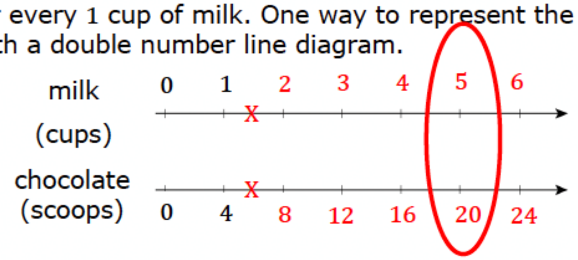
   Answers will vary. Add ¼ cup of milk.

5.2.2 Exploration: Using Double Line Diagrams to Represent Ratios

Component Overview: *Students explore representing a ratio with a double number line and use it to interpret the ratio. Consider pairing students to encourage mathematical discourse throughout this component.*

5.2.2 Exploration: Using Double Line Diagrams to Represent Ratios

Nana's chocolate milk recipe calls for 4 scoops of chocolate for every 1 cup of milk. One way to represent the recipe is with a double number line diagram.



1. Label each tick mark on the double number line diagram to represent the relationship between cups of milk and chocolate scoops.
   Shown on diagram.

2. How does the double number line show the ratio of Nana's preferred balance of milk and chocolate?
   The double line diagram shows that for every 1 cup of milk there are 4 scoops of chocolate.

3. How many scoops of chocolate need to be mixed with 5 cups of milk to maintain the same ratio of milk to chocolate from Nana's recipe?
   For 1 batch of Nana's chocolate mix, Dan needs 1 cup of milk and 4 scoops of chocolate. For 5 times that number of cups, Dan would need 5 times the number of scoops. Therefore, 4 scoops times 5 equals 20 scoops of chocolate.

4. Circle where the answer to the previous question is represented on the double number line diagram.
   Shown on diagram.

Students should use the scaffolded prompts to explore the connection between the double number line and the ratio that represents the amount of milk needed. Students could also determine the fraction by estimating.

- *What is halfway between 4 and 8?*
- *What is halfway between that? (5 scoops)*
- *What is halfway between 1 and 2?*
- *What about halfway between 1.5 and 1.5?*

5.2.2 Exploration: Using Double Line Diagrams to Represent Ratios (continued)

Each pair of numbers on the double number line diagram represents an equivalent ratio. For example, the ratio $6 : 24$ is equivalent to the ratio $2 : 8$.

5. Complete the statement to interpret these two equivalent ratios in the context of Nana's recipe.

   If 6 cups of milk are mixed with 24 scoops of chocolate, it will taste the same as mixing 2 cups of milk with 8 scoops of chocolate.

6. Recall Dan's error when mixing Nana's chocolate milk.

Nana's Recipe	What Dan Accidentally Mixed
1 cup milk 4 scoops chocolate	1 cup milk 5 scoops chocolate

   a. Sabrina said Dan's mistake can be fixed by adding 0.5 cup of milk and 1 scoop of chocolate to the glass. Discuss with your partner whether or not you think Sabrina is correct.

   b. Complete the statement.
   If Dan follows Sabrina's recommendation, the glass will contain 1.5 cups of milk and 6 scoops of chocolate.

7. Use the double number line diagram to determine whether Sabrina's recommendation will result in chocolate milk with Nana's preferred taste.

   If Sabrina's recommendation will work, mark the locations of this mixture on the double number line diagram with two X's, one on each number line.
   Shown on diagram.

Consider having students add to their reference card from Lesson 1 with an explanation on how to write ratios, interpret ratios, and represent ratios using various models. Encourage students to include at least one example of each process. Allow students to use their reference cards during the lesson.

Notice and Wonder
Mathematical discourse during this exploration should sound informal and explorative. Students have not yet connected ratios to double number lines, so as they explore this connection, encourage them to verbalize what they notice about the double number line and how it connects to ratios. Encourage students to work with their partner to verbalize their responses before putting them in writing.

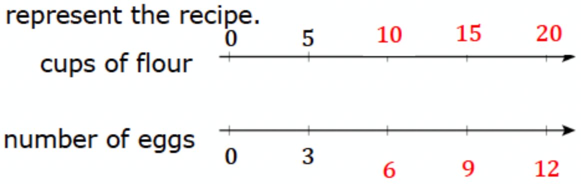
Consider asking students who finish early to represent the ratio on the double number line using the three ways they have already learned in the previous lesson. Challenge students to find other equivalent ratios for Nana's chocolate milk recipe. What if you wanted to make chocolate milk for the whole class?

5.2.3 Try It: Using Double Line Diagrams to Represent and Interpret Ratios

Component Overview: *Students practice using double number lines to represent and interpret ratios. The ratios are expressed three ways: a to b, a: b, or $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams.*

5.2.3 Try It: Using Double Line Diagrams to Represent and Interpret Ratios

1. A recipe for 1 batch of cookies calls for 5 cups of flour and 3 eggs. A double line diagram is shown to represent the recipe.



a. What is the ratio of flour to eggs in 1 batch of cookies?

5 to 3, 5:3 or $\frac{5}{3}$

b. Complete the diagram to show the amount of flour and eggs needed for 2, 3, and 4 batches of cookies.

Shown on diagram.

c. How many eggs and cups of flour are used in 4 batches of cookies?

12 eggs and 20 cups of flour

d. Complete the ratio comparing cups of flour to 6 eggs.

$\boxed{10} : 6$

e. Complete the ratio of eggs to 15 cups of flour.

$\dfrac{\boxed{9}}{15}$

MA.K12.MTR.2.1: Demonstrate understanding by representing problems in multiple ways.
Students see ratios represented multiple ways. The ratios are expressed three ways: a to b, a: b, or $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Discussing the answers with the students and requiring them to justify their responses will help them draw connections among these representations and synthesize their understanding.

Engage the class in a discussion following the completion of the questions. Draw connections between the ways ratios are expressed and how they can be represented visually. Have students verbalize how they can use the ratio in one batch of cookies to determine the ratios for multiple batches of cookies to solidify understanding.

5.2.4 Guided Instruction: Using Tape Diagrams to Represent Ratios

Component Overview: *Students use tape diagrams to represent ratios. Guide students through this component. Engage the class in a discussion to solidify the connection between the tape diagrams and ratios.*

5.2.4 Guided Instruction: Using Tape Diagrams to Represent Ratios

A tape diagram is another way to represent ratios. All the parts of the diagram that are the same size have the same value, similar to increments on a number line.

The tape diagram shown represents the ratio of cats to dogs at a local animal shelter.

1. Complete the statements using the tape diagram.

   a. The ratio of dogs to cats is $\frac{5}{4}$.

   b. For every __4__ cats at the shelter, there are __5__ dogs.

2. If there are 27 cats and dogs at the animal shelter in all, and the ratio of cats to dogs is 4:5, the same tape diagram can be used to represent the relationship.

   a. What is the value of each small rectangle in the diagram given this new information? 3
   b. Label each small rectangle with this value. *Shown on diagram.*
   c. How many dogs are at the animal shelter? 15
   d. How many cats are at the animal shelter? 12
   e. Complete the statement. The ratio of cats to dogs at the shelter, 4:5, is equivalent to the ratio 12 to 15.

Consider using a kinesthetic representation of the number line on a clothesline. Students can place and adjust numbers on folded paper/cardstock on the clothesline.

Students may struggle to understand how the same tape diagram can be used to represent a ratio comparing different numbers. Ask the following questions to clarify up misconceptions:

- *If the ratio of cats to dogs stays 4:5, how can you use the tape diagram to show this?*
- *How many cats and dogs were at the shelter in the first example?*
- *How can you determine this? How can you visually represent the new total of animals on the tape diagram? Consider how you can get from 9 to 27.*

Students may struggle to conceptually understand and articulate why the ratios are equivalent, even if they are able to represent them correctly on the tape diagram.

Utilize inquiry ("How do you know?") for parts a-e and talk moves ("Restate in your words what your classmate said.") to formalize their conceptual understanding before they independently practice in 5.2.5 Your Turn!.

5.2.5 Your Turn!

Component Overview: *Students work independently to represent and interpret ratios using double number lines and tape diagrams. Students then connect equivalent ratios to multiplication and division equations more formally, creating equations to find equivalent ratios.*

5.2.5 Your Turn!

1. Mr. Floyd is planning a field trip for his class to the Cade Museum for Creativity and Invention in Gainesville, Florida. To meet the museum's recommended ratio of chaperones to students, Mr. Floyd needs to have 1 adult chaperone for every 9 students. The tape diagram represents the ratio of adults to students needed.

number of adults	8								
number of students	8	8	8	8	8	8	8	8	8

   a. What is the ratio of students to chaperones required by the museum?

   Any form of ratio representation is accepted:
   9 *to* $1, 9{:}1$ *or* $\frac{9}{1}$

   There are 72 students who will attend the trip from Mr. Floyd's classes. Use the tape diagram to determine how many chaperones will be needed.

   b. Label each small rectangle with the number of people it represents.
   Shown on diagram.

   c. Complete the statement.

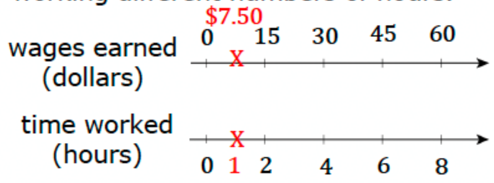
   For all 72 students to attend, 8 chaperones are needed.

Students may struggle to use the tape diagram to represent the new ratio. Encourage students to refer back to the previous component and consider how they represented equivalent ratios in the previous examples.

- *If the ratio of students to teachers is 9:1, how many teachers will be needed for 72 students?*
- *How can you determine what this equivalent ratio will be?*
- *How did you determine the equivalent ratio in the previous example?*

5.2.5 Your Turn! (continued)

2. A cashier worked an 8-hour day and earned $60.00. The double number line represents the amount she earned for working different numbers of hours.



a. Determine the amount of money she earned in 1 hour. Show your work and label it on the double number line.
Shown on diagram. $60/8 = \$7.50$

b. How much money does the cashier earn if she works 3 hours?
$\$7.50\ x\ 3 = \$22.50$

Encourage students to think through how they can determine each amount and how they can show their work.

5.2.6 Wrap-Up: Ticket Out the Door

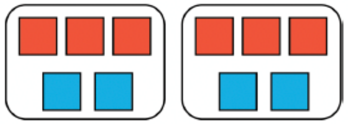
Component Overview: *Students use a tape diagram to create a double number line and interpret a ratio.*

5.2.6 Wrap-Up: Ticket Out the Door

1. The representation shows the amount of red and blue paint that makes 2 batches of a particular shade of purple paint called lilac.

cups of red paint

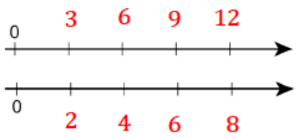
cups of blue paint



a. On the double number line, label the tick marks to represent amounts of red and blue paint used to make batches of this shade of purple paint.

cups of red paint

cups of blue paint



b. How many batches are made with 12 cups of red paint?
4 batches

c. How much red paint should be mixed with 6 cups of blue paint to make lilac paint?
9 cups of red paint

d. Complete the statement.

To make this shade of lilac paint, mix 2 cups of blue paint for every 3 cups of red.

Students may struggle using a double number line but correctly apply ratios and identify solutions. There is not one "correct" way to represent ratios using various models as described in the benchmark.

For students who correctly identify parts b-d but struggle with part a, consider offering them the opportunity to represent their solution using a model of their choice.

5.3 Exploring Rates

Lesson Introduction

Benchmark	MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.	Learning Targets	• I can use ratios and rates to describe relationships between quantities. • I can write a ratio as a unit rate.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

Essential Questions	• How do you use ratios and rates to describe relationships between quantities? • How do you write a ratio as a unit rate?
Lesson Narrative	In this lesson, students deepen their understanding of ratios. Students begin by representing ratios in different ways and connecting them to real-world contexts, and then connecting this understanding to rates and then unit rates. These new concepts are then applied to more real-world scenarios, first with units of measurements such as miles, and then extended to more abstract scenarios. The lesson culminates in a self-assessment to identify areas of struggle.

Component Summary				
	5.3.1 Warm-Up (~5 min.)	Students decide and justify which bag of chips they would rather share with a friend *(SE p. 184)*	5.3.4 Guided Practice (5 min.)	Practice using ratios and rates *(SE p. 186)*
	5.3.2 Collaboration (20 min.)	Work with a partner to express ratios as a to b, $a{:}b$, and $\frac{a}{b}$ *(SE p. 184)*	5.3.5 Your Turn! (5 min.)	Write unit rates from real-world scenarios *(SE p. 187)*
	5.3.3 Guided Instruction (10 min.)	Express ratios as rates and unit rates *(SE p. 185)*	5.3.6 Wrap-Up (~5 min.)	Red, Yellow, Green Light self-reflection *(SE p. 187)*

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Rate • Unit rate Additional Terms: N/A	• (Optional) Note cards • (Optional) Chart paper • (Optional) Markers	• Create an anchor chart. • Consider bringing in some measuring cups and/or bags of chips and containers of sauce to help students visualize the size of the quantities they are working with in the problems.

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>

Students may forget to simplify ratios with division before multiplying to find the unknown unit.
Students may invert ratios when trying to find an equivalent rate.
Students may struggle to understand differences between fractions and ratios and what they represent.

</td><td>

Consider creating an anchor chart and allowing students to add to their reference cards with instructions and examples to aid them in using ratios and rates to describe relationships between quantities and writing a ratio as a unit rate.
Remind students to label the units in the quantities and then to keep the unit placement the same in the second ratio. This will help avoid mistakes.

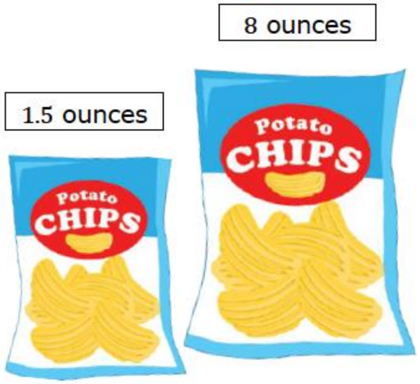
</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
5.3.1 Warm-Up provides an engaging entry point for students to debate with a friend. Use the “Would You Rather?” prompt to position students as claim-makers, using evidence with at least one number in their responses to prove who might have the better answer. There is no one right answer, so encourage creativity in responses.</td><td>MA.K12.MTR.2.1: Multiple Representations
Throughout the lesson students see ratios represented multiple ways. The ratios are expressed three ways: a to b, $a\!:\!b$, and $\frac{a}{b}$. The ratios are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout the lesson.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The Wrap-Up asks students to write which topics they feel stuck on, are starting to get the hang of, and feel confident in. Consider allowing students to choose whether to record themselves explaining the answer verbally, drawing and labeling a diagram, or providing an example.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.3.1 Warm-Up: Would You Rather?

Component Overview: *Students compare and justify which bag of chips they would rather share with a specified number of friends.*

5.3.1 Warm-Up: Would You Rather?

1. Would you rather share the small bag of chips with 1 friend or the large bag of chips with 8 friends? Explain your reasoning using mathematics.
   Answers will vary.
   I would rather share the bigger bag since it would give more chips per person than the smaller bag.



Think-Pair-Share
Ensure that students realize two people (one friend and you) are sharing the chips in the first scenario, and nine are sharing the chips in the second scenario. Consider positioning students to discuss question prompts before or instead of writing their answers. Use sentence stems to encourage conversation such as, "I would rather share the … bag with one friend because…" and "I would rather share the … bag with eight friends because…"

5.3.2 Collaboration: Expressing Ratios in Different Ways

Component Overview: *Students work with a partner or small group to express ratios as a to b, a: b, and $\frac{a}{b}$. Once students have completed the two questions, share answers as a class and discuss how each group arrived at their response.*

5.3.2 Collaboration: Expressing Ratios in Different Ways

Ratios are often expressed in the real world as an amount of "something per something." For example, in setting the world record, Joey Chestnut ate 7.4 hot dogs per minute.

Discuss each of the following scenarios with your partner and work together to answer the questions.

1. A recipe for marinara sauce requires 56 ounces (oz) of whole peeled tomatoes. The recipe yields 8 servings of marinara sauce.

   a. What is the ratio of tomatoes to sauce?
   56 ounces of tomatoes to 8 servings of sauce.
   56 to 8, 56:8 or $\frac{56}{8}$

   b. Complete the tape diagram to represent the ratio of tomatoes to sauce.

tomatoes (ounces)	7	7	7	7	7	7	7	7
sauce (servings)	1	1	1	1	1	1	1	1

   c. How many ounces of whole peeled tomatoes are in 1 serving of marinara sauce? *7 oz.*

   d. Complete the ratio.

$$\frac{7 \text{ oz of tomatoes}}{1 \text{ serving}}$$

Students may discuss whether each box should have 7 ounces or 8 servings. For students who are unsure, consider prompting questions.

- *Why are there 8 boxes? What do they represent?*
- *How would you determine the equal amount in each of the 8 servings for a total of 56 ounces?*
- *What if there were 14 ounces in 2 servings? How many ounces would be in each serving?*

5.3.2 Collaboration: Expressing Ratios in Different Ways (continued)

2. In 2012, there were 982 farms in Hardee County, Florida. The total amount of farmland was 273,978 acres.

   a. Write the ratio of acres : farms.
   273,978:982, *or* $\frac{273{,}978 \text{ acres}}{982 \text{ farms}}$, *or* 273,978 *to* 982

   b. Complete the statement.

   The total amount of farmland in Hardee County is split up among all the farms, so to determine the average number of acres for 1 farm,

   - add 982 to 273,978.
   - subtract 982 from 273,978.
   - multiply 273,978 by 982.
   - divide 273,978 by 982.

   c. Complete the ratio.

   $$\frac{279 \; acres}{1 \; farm}$$

   d. Discuss with your partner whether you think this number of acres per 1 farm means that every farm in Hardee County was that size in 2012. Then, write a brief summary of your reasoning.

   Answers will vary. No, because not everyone will purchase the same amount of land. Some people may be able to afford 1 acre whereas others may be able to afford 279 acres.

Students should be concluding that they must divide to determine the ratios for each of these questions. Listen for students who are making the connection that because ratios are multiplicative, this means they can divide to determine the ratio per 1. Highlight this thinking before moving onto the next component, as it will prime students for the discussion of unit rates.

5.3.3 Guided Instruction: Understanding Rates

Component Overview: *This Guided Instruction introduces students to rates and unit rates.*

5.3.3 Guided Instruction: Understanding Rates

Each of the ratios discussed so far in this lesson are also ***rates***.

A ***rate*** is a ratio that compares two quantities of different units.

A ratio is a comparison of two numbers, so all rates can also be expressed using ratios. Rates are often used to express how long it takes to do something.

1. Consider a familiar scenario involving a rate. Tyler ran 2 miles in 30 minutes.

   a. Draw a tape diagram for the ratio that represents Tyler's run.

Miles	1 mile	1 mile
Minutes	15 minutes	15 minutes

   b. Complete the ratios. Then, complete each statement to interpret the ratios.

   $\frac{\frac{1}{15}\ miles}{1\ minute}$ $\frac{15\ minutes}{1\ mile}$

   Tyler runs $\frac{1}{15}$ miles for every 1 minute.
   Tyler runs 15 minutes for every 1 mile.

   c. Discuss with your partner how to use one of the rates to determine how many miles Tyler runs in an hour. Then, complete the statement.
   Tyler runs 4 miles per hour.

Each of the rates used to describe Tyler's run is a ***unit rate***.

A ***unit rate*** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

Turn and Talk
After the definitions are read out loud, have students participate in a Turn and Talk routine to synthesize each definition. Students should not be reading the definition or parroting what is already in the box, but instead using their own words to demonstrate understanding. If students are struggling to define in their own words, encourage them to come up with an example of a rate and unit rate.

Students may struggle to understand the difference between rate and unit rates or may not comprehend the significance of being able to calculate both. Encourage students to consider why knowing the unit rate may be more helpful in certain scenarios.
How is a unit rate different than a rate?
Why is it important to know the unit rate of comparison?
When might it be helpful to use the unit rate?

5.3.4 Guided Practice: Using Ratios and Rates

Component Overview: *Students represent the relationship between various ingredients and calories on a nutrition label using ratios. Consider partnering students in pairs during this component.*

5.3.4 Guided Practice: Using Ratios and Rates

1. A nutritional label for raspberry ice cream is shown.

Nutrition Facts

3 servings per container

Serving size **2/3 Cup (123g)**

	Per Serving	Per Container	
Calories		**1000**	
			% Daily Value*
Total Fat		58g	74%
Saturated Fat		36g	180%
Trans Fat		1.5g	
Cholesterol		385mg	128%
Sodium		550mg	24%
Total Carbohydrate		106g	39%
Dietary Fiber		0g	0%
Total Sugars		88g	
Includes Added Sugars		72g	144%
Protein		16g	
Vitamin D		3mcg	15%
Calcium		405mg	30%
Iron		2mg	10%
Potassium		532mg	10%

*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

a. What is the ratio of calories to servings?
$1{,}000 : 3$

b. What is the ratio of total carbohydrates in the container to the number of calories in the container?
$106 : 1{,}000$

c. What is the ratio of added sugars to the number of servings?

$72 : 3$

d. Complete the unit rate to express the number of added sugars per serving.
$\frac{24\ grams}{1\ serving}$

Consider bringing in a variety of snacks or have one student per group bring in their group's favorite snack to share while working. Students can demonstrate their understanding using their favorite snack in substitution of or in addition to this label provided.

Challenge students who finish early to find the unit rates of each relationship.

5.3.5 Your Turn!

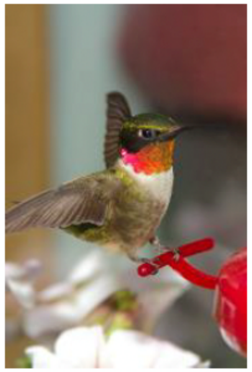
Component Overview: *Students work independently to write unit rates from real-world scenarios.*

5.3.5 Your Turn!

1. Tracy McGrady is a former NBA basketball player from Bartow, Florida. He played in 938 games in the NBA, scoring a total of 18,381 points. Complete the ratio to express the number of points he scored per game as a unit rate. Round the number of points to the nearest tenth.

   $$\frac{19.6\ points}{1\ game}$$

2. The ruby-throated hummingbird is the most common hummingbird found in Florida. This bird can flap its wings 1,590 times in 30 seconds. Complete the ratio to express how fast the ruby-throated hummingbird flaps its wings per second.

   53 *wing flaps* : 1 *second*

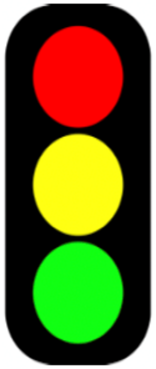
Remind students to label the units that correspond to each quantity in the ratio and then to keep the unit placement the same in the second ratio. This will help avoid mistakes.
What does "per" mean?
How did you find the points per game?
How did you find the wing flaps per second?

5.3.6 Wrap-Up: Reflection

Component Overview: *Students use a stoplight reflection to consider which topics they feel stuck on, are starting to get the hang of, and feel confident with.*

5.3.6 Wrap-Up: Reflection

1. Complete the stoplight reflection to share how you are feeling about the concepts learned in today's lesson.



I still feel stuck about how to ...
Answers will vary. If students are completely stuck they may choose this option.

I'm starting to get the hang of ...
Answers will vary. If students understand it but need more practice they may choose this option.

I feel confident and ready to go with ...
Answers will vary. If students are completely confident in this topic they may choose this option.

For students who might struggle verbalizing their understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can be used for students to reflect.

5.4 Exploring Tables of Equivalent Ratios

Lesson Introduction

<table>
<tr><td>Benchmarks</td><td colspan="2">MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.
MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.</td><td>Learning Target</td><td colspan="2">• I can build tables of equivalent ratios incorporating an understanding of unit rate.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="2">MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.</td><td colspan="3">MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.</td></tr>
<tr><td>Essential Questions</td><td colspan="5">• How do you use ratios to describe relationships between quantities?
• How do you use unit rates to describe relationships between quantities?
• How do you write a ratio as a unit rate?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="5">In this lesson, students build on their understanding of ratios and unit rates through an exploration of the number of balloons it will take to float objects of various weights. Students begin by determining the unit rate of balloons to weight in grams and then apply this rate to different weights in a table. Students deepen their understanding of the relationship between ratios and unit rates by using ratios in a table to determine the unit rate for different objects, noticing the unit rate will be the same for each, indicating that the ratios are equivalent. This lesson may also be completed using Desmos Activity Builder. Teachers can choose to facilitate the lesson through the Desmos activity or the paper-based version in the workbook if that is not an option.</td></tr>
<tr><td rowspan="2">Component Summary</td><td>5.4.1 Warm-Up (~5 min.)</td><td>Students watch a video in which a man floats using balloons and ask a question they have (SE p. 188)</td><td>5.4.3 Guided Practice (20-25 min.)</td><td colspan="2">Use tables to explore equivalent ratios and write unit rates (SE p. 190)</td></tr>
<tr><td>5.4.2 Exploration (15-20 min.)</td><td>Explore how many balloons it will take objects with different weights to float and record the data in a table (SE p. 189)</td><td>5.4.4 Wrap-Up (~5 min.)</td><td colspan="2">Students describe how to calculate how many balloons it will take a friend to float (SE p. 192)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td><u>Key Terms</u>: N/A

<u>Additional Terms</u>: N/A</td><td colspan="2">• (Optional) Desmos Activity Builder
• Balloon Float Video for 5.4.1 Warm-Up</td><td colspan="2">• Consider making an anchor chart.</td></tr>
</table>

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>• In 5.4.3 Guided Practice, students may not realize that the third column asks for grams per balloon, so it should be divided by balloons or grams-to-balloon rate.
• Students may forget that one quantity in a unit rate is 1 unit.
• Students may forget to simplify ratios with division before multiplying to find the unknown unit.
• Students may invert ratios when trying to find an equivalent rate.</td><td>• Remind students to label the units when comparing two quantities in a ratio and then to keep the unit placement the same in the second ratio. This will help avoid mistakes.
• Remind students that “per balloon” means “per one balloon.” The same applies to the grams-to-pound oats example.
• Remind students that fractions always express a part-to-whole comparison, but ratios can express a part-to-whole comparison or a part-to-part comparison, which can be written as a to b, $a{:}b$, and $\frac{a}{b}$.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
5.4.1 Warm-Up and 5.4.2 Exploration are structured with a Notice and Wonder routine. Students are asked what they notice and wonder about a video of a man using balloons to float in the air.</td><td>MA.K12.MTR.5.1: Patterns and Structure
In 5.4.2 Exploration students look at the number of balloons needed to lift objects of different weights. Then, students use an equation to generalize the relationship between the number of balloons and the weight of the object in grams.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Digitally engage students with the pre-created Desmos Activity Builder for this lesson. Each component in this lesson is modeled and reflected in this activity, with opportunities for students to interact as they make predictions on objects floating and prove their conjectures mathematically.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.4.1 Warm-Up: Balloon Lift

Component Overview: *Play the video of the man floating with the balloons and then have students participate in a "Notice and Wonder" routine to discuss what they noticed in the video and what they wonder. Students record their answers to question 1.*

5.4.1 Warm-Up: Balloon Lift

In 2015, a Canadian man attached enough balloons to his lawn chair to float up into the sky.

1. Write a question, a wonder, or a comment about this scenario to be used during a class discussion.

   Answers will vary. How many balloons are there? Is this person going to stay floating forever? How long was he in the air? How long would balloons keep me in the air?

Notice and Wonder
Observe student responses either by circulating to take note or by looking at the responses submitted via the Desmos activity.
In the whole-class discussion following the Warm-Up, highlight unique questions for the class, both mathematical and non-mathematical. This is a great time to introduce the focus of the activity, which is to answer the question: "How many balloons will it take to make the person float?"

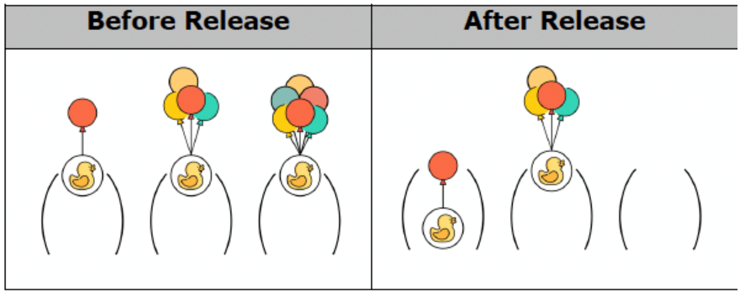
5.4.2 Exploration: Make It Float

Component Overview: *Students work in pairs to continue to explore the relationship between the number of balloons needed for an object to float and the object's weight in grams. Students then use two tables to make predictions about the number of balloons it will take for certain objects to float based on their weight in grams and explain their predictions. Review the second table as a class and encourage students to verify their responses and clear up any misconceptions before moving onto 5.4.3 Guided Practice.*

5.4.2 Exploration: Make it Float

Flying high with balloons is dangerous! This activity will focus on determining how many balloons it takes to float various objects.

The images below show different numbers of balloons attached to rubber ducks of equal weight. The first image shows the ducks before they are released from the hold. The second image shows what happens after they are released.



1. Write one thing you know, one thing you wonder, and one thing you learned after looking at the figures above.

   a. One thing I know is …
   Answers will vary. The duck with one balloon fell to the ground.

   b. One thing I wonder is …
   Answers will vary. How many balloons does it take to start rising? Is it different for different weights?

   c. One thing I learned is …
   Answers will vary. It takes 4 balloons to keep a duck floating.

To bring awareness to the importance of the ratio of balloons to weight, pose the following questions as students are working:

- *What happens if there are not enough balloons?*
- *What happens if there are too many balloons?*
- *What happens if the correct number of balloons are attached to the object?*

Students may not understand that the third duck floated away because it was attached to too many balloons. Be sure students understand this, so they make the connection that four is the necessary number of balloons to float.

5.4.2 Exploration: Make It Float (continued)

2. It took 4 balloons to make the duck float. The duck weighs 56 grams. If each balloon lifts the same amount, how many grams would float with 1 balloon?
   $14\ g$

3. How many balloons will it take to float a toy bear that weighs 350 grams?

   a. Write your prediction in the table.

Object	Weight (grams)	Number of Balloons
Sandbag	14	1
Duck	56	4
Bear	350	25

   b. Explain the reasoning for your prediction.
   Dividing the weight of the duck, 56g, by 4 balloons, the result is that 14g will float with one balloon.
   With this, dividing the 350g of the bear by 14, the result is that it will take 25 balloons to make the bear float.

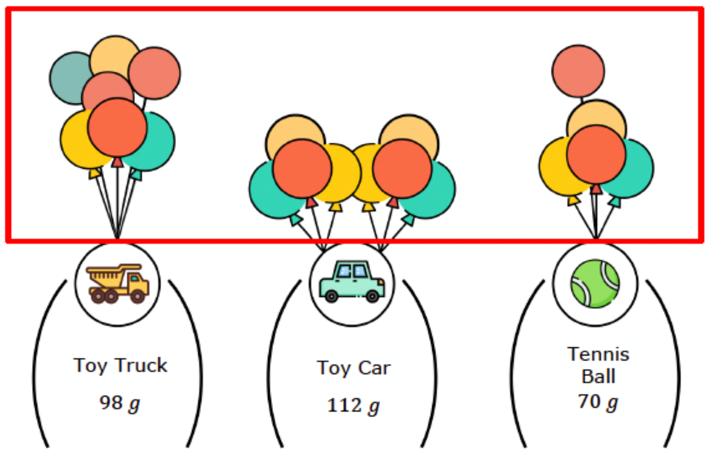
Think-Pair-Share
Consider structuring this component with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.

5.4.2 Exploration: Make It Float (continued)

4. Complete the table for the number of balloons needed to float each object based on its weight.

Object	Weight (grams)	Number of Balloons
Sandbag	14	1
Duck	56	4
Bear	350	25
Toy Truck	98	7
Toy Car	112	8
Tennis Ball	70	5

5. Draw the balloons for each object that would make it float.



Challenge advanced learners to find the weight of classroom objects (using a scale borrowed from the science teacher) and determine how many balloons would be needed to make it float. If a scale is not available, consider providing estimated weights for students to work with instead.

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

Component Overview: *Students work in pairs to complete the table to show the unit weight lifted by one balloon. Engage the class in a "Notice and Wonder" routine after completing the first table, then have students continue to work in pairs to determine how many balloons can lift the average adult. Students then complete another table to determine how many of a different type of balloon are needed to lift objects based on weight, using the table to determine how many of the new balloons are needed to lift an average adult.*

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

We have looked at the number of balloons needed to lift an object of a certain weight.

1. Use the relationship between the number of balloons, B, and the weight of the object in grams, g, to write the unit rate of $\frac{\text{weight in grams}}{\text{number of balloons}}$.

   a. Complete the table to show the unit weight lifted by a balloon.

Object	Weight (grams)	Number of Balloons	Unit Rate (grams per balloon, $\frac{g}{B}$)
Sandbag	14	1	$\frac{g}{B} = \boxed{14}$ *grams per balloon*
Duck	56	4	$\frac{56}{4} = 14$ *grams per balloon*
Bear	350	25	$\frac{350}{25} = 14$ *grams per balloon*
Toy Truck	98	7	$\frac{98}{7} = 14$ *grams per balloon*
Toy Car	112	8	$\frac{112}{8} = 14$ *grams per balloon*
Tennis Ball	70	5	$\frac{70}{5} = 14$ *grams per balloon*

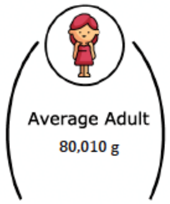
Students may struggle to set up the ratio to find the unit rate in this table because there is not a visual setup with variables, as there was in question 1. Consider guiding students with the following questions if you notice they are having trouble setting up the ratio to determine the unit rate:

- *How do you determine the unit rate?*
- *If you are determining the weight per balloon, what does "per" mean in this instance?*
- *Using the answer to the previous question, where should the weight in pounds be represented in the ratio?*
- *Where should the number of balloons be represented in the ratio?*

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables (continued)

b. What do you notice after completing the table?
Answers will vary. The unit rate is the same for all the objects.

2. A diagram of an average adult whose weight is 80,010 grams (g) is shown.



a. Discuss with your partner how to use the information from the previous table to determine how many balloons it would take to float the person.

b. Complete the statement.
It would take 5,715 balloons to float the person.

3. Weather balloons would be better to use in this situation. Unlike regular balloons, weather balloons are designed to lift heavier objects.

The partially completed table shows the weight in pounds (lbs) of four objects and the number of weather balloons needed to float each object.

a. Complete the table with the weight, the number of weather balloons, and unit rate for each object.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Table	34	4	$\frac{34}{4} = 8.5$ *lbs. per balloon*
Flowerpot	17	2	$\frac{17}{2} = 8.5$ *lbs. per balloon*
Bookshelf	51	6	$\frac{51}{6} = 8.5$ *lbs. per balloon*
Armchair	93.5	11	$\frac{93.5}{11} = 8.5$ *lbs. per balloon*

Notice and Wonder
After students complete the table, implement a "Notice and Wonder" routine for question 1b.

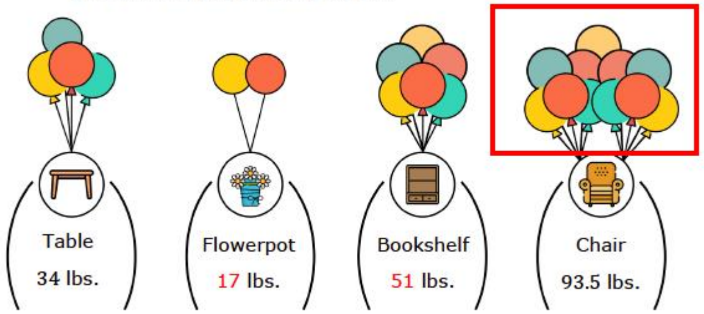
- *First, have students consider the table and tell a partner what they notice. Consider recording common observations on chart paper or a white board for the class to see.*
- *Then, have students tell their partner what the table makes them wonder.*
- *Share out as a class, and record on chart paper or a white board.*

Guide students toward the conclusion that it takes one balloon to float exactly 14 grams if this has not already been discussed, and have students consider how they can use the unit rate to determine how many balloons are needed to float other objects.

The tables provide a visual entry point. Allowing students to discuss the problems provides an auditory entry point. If students have not mastered their multiplication and division facts, let them use a calculator to determine the unit rate. The calculator will allow students to focus on mastering the learning goals of this lesson.

5.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables (continued)

b. Draw the weather balloons needed to make the chair float and write the weight of each object determined from the table.



4. Consider the figure shown.

How many weather balloons would it take to float the person? Write your answer in the table.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Person	204	24	$\frac{204}{24} = 8.5$ *lbs. per balloon*

The tables provide a visual entry point. Allowing students to discuss the problems provides an auditory entry point. If students have not mastered their multiplication and division facts, let them use a calculator to determine the unit rate. The calculator will allow students to focus on mastering the learning goals of this lesson.

Consider challenging students who may finish early to determine how many balloons it would take to float a classroom object or a pet at home.

5.4.4 Wrap-Up: Cool Down

Component Overview: *Students describe how to calculate how many balloons it will take a friend to float. The Wrap-Up provides students with an opportunity to demonstrate their understanding of the learning goal of the lesson.*

5.4.4 Wrap-Up: Cool Down

1. Your friend wants to know how to calculate the number of balloons she would need to float.

   Answer the questions in the space provided in the table.

What information do you need to know?	*Her weight in pounds and the unit rate (pounds per balloon)*
What instructions would you give her about how to calculate the number of balloons she would need to float?	*To calculate the number of balloons that she would need to float, she would divide her weight by the unit rate of 8.5 lbs per balloon for weather balloons.*

Do you have extra time and access to tools like a scale and balloons?
Consider an extension project in which students choose classroom objects, make a conjecture for how many balloons it would take to make it float, and then mathematically prove their answers.

5.5 Connecting Ratios and Rates

Lesson Introduction

Benchmark	MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.	Learning Target	• I can make connections between ratios, rates, and unit rates.
Benchmark Coherence	**Building On** MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.	**Working Toward** N/A	
Essential Questions	• What strategies can you use to calculate the unit rate and associated ratios? • What two mathematical operations are used to convert measurement in one unit to measurement in another unit? • If work is being done at a constant rate by one person, and a different constant rate by another person, how do you convert them to compare them directly?		
Lesson Narrative	Students explore how to scale recipes up and down using ratios, multiplication, and division, connecting ratios more formally to unit rates. Students then build on this understanding as they use ratios and rates to describe relationships in the context of speed.		

Component Summary

5.5.1 Warm-Up (~5 min.)	Analyze examples of mathematics in nature *(SE p. 193)*	5.5.4 Try It (10 min.)	Practice comparing the ratio, rate, and unit rates of a slug and a snail *(SE p. 196)*
5.5.2 Guided Instruction (10 min.)	Solve ratio and rate problems using double number lines and ratio tables *(SE p. 194)*	5.5.5 Your Turn! (10 min.)	Practice solving real-world problems involving ratios, rates, and unit rates using a double number line and a ratio table *(SE p. 197)*
5.5.3 Collaboration (10 min.)	Compare the ratio, rate, and unit rates of two runners *(SE p. 195)*	5.5.6 Wrap-Up (~5 min.)	Student reflection on which topics they are still unsure and plan to improve understanding of these topics *(SE p. 197)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: N/A Additional Terms: N/A	• (Optional) Chart paper • (Optional) Markers	• Warm-Up 5.5.1 contains a video to accompany the question prompts.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may think that the object that traveled the farthest is the fastest (if they do not consider how much time it took). • Students may struggle to make connections between tape diagrams, double number lines, and ratio tables, believing them to be different quantities rather than the same quantity represented in different ways.	• Remind students that a rate is a comparison of two quantities that have different units of measure and that the ratio is not complete unless it has units. • Consider displaying the ratio table horizontally to help students connect it to a tape diagram or double number line. • Create an anchor chart and allow students to create reference cards with instructions and examples to aid them in applying ratio relationships to solve real-world problems.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Consider structuring 5.5.3 Collaboration with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.	**MA.K12.MTR.2.1: Multiple Representations** Students apply ratio relationships to solve real-world problems. Across the lesson components, the ratios are represented in ratio tables and double number lines. Students make connections among these representations and determine the most efficient representation to use and how to use it to solve problems.
	Consider using double number lines and equivalent ratio tables to help students visualize the ratios in the 5.5.4 Try It and 5.5.5 Your Turn! components of the lesson. Consider having students represent the problem using both representations as an additional scaffold, if needed.	

Lesson Notes:

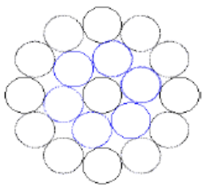
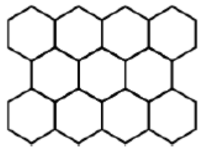
5.5.1 Warm-Up: Why do Honeybees Love Hexagons?

Component Overview: *Students justify why a hexagonal shape allows bees to store the most honey using the least amount of wax and share another example of mathematics in nature.*

5.5.1 Warm-Up: Why do Honeybees Love Hexagons?

Honeybees are actually amazing mathematicians. Scientists claim they can even calculate angles. They also live inside one of the most mathematically efficient architectural designs there is: the beehive. It takes a lot of work for bees to produce wax. The hexagonal shape of the cells in a honeycomb allows bees to store the most amount of honey while using the least amount of wax. This design likely evolved over time.

Consider the shapes used in the designs shown.

Triangular Cells	Circular Cells	Hexagonal Cells
		

1. How do the designs demonstrate that the hexagonal shape allows bees to store the most amount of honey using the least amount of wax?
   Answers will vary. The hexagonal shape allows bees to store the most amount of honey using the least amount of wax because there are no spaces in the pattern which allow for no wasted space as is the case of the circular shape.

2. This is an example of mathematics in nature. What is another example of mathematics in nature you can think of?
   Answers will vary. Repeated patterns on leaves are an example of mathematics in nature.

Consider allowing students to discuss question prompts rather than writing their answers for classes that engage better using routines like Turn and Talk or Think-Pair-Share.
Use sentence stems to encourage conversation such as: "From least to greatest, the design with the greatest area (most honey) is…" and "From least to greatest, the design with the least perimeter (least wax) is …" "Therefore, the best design (most honey and least wax) is…"

5.5.2 Guided Instruction: Connecting Ratios and Rates

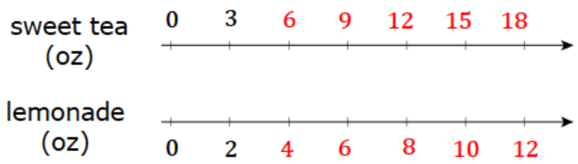
Component Overview: *Students use a double number line and corresponding table to represent a ratio used to make a drink mixture, then use what they know about unit rate to determine which operations to use when scaling the recipe up and down. Engage the class in a discussion to highlight questions 3-5 and solidify understanding for how to find equivalent ratios.*

5.5.2 Guided Instruction: Connecting Ratios and Rates

A given ratio or rate can be used to generate many equivalent ratios, which can be helpful in solving problems. Double number lines, tape diagrams, mixture charts, and ratio tables are useful tools for finding equivalent ratios.

1. Noah's favorite drink is a mixture of sweet tea and lemonade. He uses the ratio of 3 ounces (oz) of sweet tea to 2 oz of lemonade to make it.

   A double number line diagram is shown to represent the situation. Complete the double number line diagram.



A table can also be used to represent this relationship.

2. Complete the table to show how many ounces of each beverage are needed to make different-sized batches of Noah's favorite drink.

Sweet Tea (oz)	Lemonade (oz)
3	2
21	14
15	10
4.5	3
48	32
1.5	1

Recipes are scaled up to make larger batches with the same taste. They can also be scaled down to make smaller batches with the same taste. Equivalent ratios can be used to determine the amount of ingredients needed when scaling a recipe up or down.

This Guided Instruction guides students in how to apply the knowledge they have learned in this unit to connect the double number line and table representations. Students may not need this to be modeled, as it is practice from skills they have been building in previous lessons.
Bring the class together to complete questions 3-5 to highlight the ways in which equivalent ratios are created before students continue working with ratios and unit rates in tables. Ensure students can articulate the ways in which equivalent ratios can be determined.

3. When scaling a recipe up, such as to find how many ounces of sweet tea to mix with 32 ounces of lemonade, which operation is used?
   Multiplication

4. When scaling a recipe down, such as to find how many ounces of lemonade to mix with 1.5 ounces of sweet tea, which operation is used?
   Division

5. William noticed he could use similar thinking with the double number line diagram. He drew an arrow from 3 to 15 on the top number line and another arrow from 2 to 10 on the bottom number line. Complete the statement to describe what he saw.
   William noticed the ratio $\frac{3}{2}$ can be used to find the ratio $\frac{15}{10}$ by
   - ○ adding 12 to both parts of the ratio.
   - ○ multiplying both parts of the ratio by 5.
   - ○ dividing both parts of the ratio by 5.

Equivalent ratios can be found by multiplying or dividing each part of a ratio by the same value.

6. Show how the ratios 3:2 and 48:32 from the table are equivalent, using either multiplication or division.
   $\frac{3}{2} \times \frac{16}{16} = \frac{48}{32}$ or $\frac{48}{32} \div \frac{16}{16} = \frac{3}{2}$

7. Shay noticed the table of Noah's favorite drink included a unit rate. Circle the unit rate in the table. Then, complete the statement to explain its meaning in context.
   Shown in table.

   This unit rate means there are 1.5 ounces of *sweet tea* in the drink for each ounce of *lemonade*.

Students may confuse when to multiply to find the equivalent ratio and when to divide, or they may not understand the vocabulary of "scaling up" and "scaling down." Have students consider what is happening when they are creating more/less of a recipe, and what operation can help them accomplish that. It will be helpful to connect this to a visual for students who may struggle.
In the discussion, guide students to make connections between "equal groups" and multiplication and division.

- *What is happening when you scale up a recipe?*
- *Will there be more or less of the drink?*
- *What operation can be used to create more?*
- *What happens when you scale down a recipe?*
- *What operation can be used to create less?*
- *How can you show this using a double number line?*

5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Component Overview: *Students work with a partner to compare the ratio, rate, and unit rate of two runners. Consider structuring this section with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning.*

5.5.3 Collaboration: Considering Ratios and Rates in the Context of Speed

Mai and Jada each ran on a treadmill. The treadmill display shows the distance, in miles, each person ran and the amount of time it took them, in minutes and seconds.

Mai's Treadmill Display	Jada's Treadmill Display
	

1. Discuss with your partner the similarities and differences between Mai's and Jada's workouts. Write them in the table below.
   Answers will vary.

Similarities	Differences
Distance $= 3.0$ *miles*	*Minutes* $= 24{:}00$ *and* $30{:}00$
Incline $= 10$	*Calories* $= 481$ *and* 411
Level $= 12$	*Pace* $= 8{:}00$ *and* $10{:}00$
Pulse $= 125$	

2. Express each person's workout as a ratio of distance to time, including units.

	Ratio of distance to time
Mai's Workout	$3\ miles : 24\ minutes$
Jada's Workout	$3\ miles : 30\ minutes$

Think-Pair-Share
Consider structuring questions 1-3 with the Think-Pair-Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.

3. Each person ran at a constant rate by setting the speed on their treadmill. Discuss with your partner who ran at a faster rate using the ratios.

   The ratio of distance to time for Mai is $3 : 24$. The ratio of distance to time for Jada is $3 : 30$. It took Jada 30 minutes to run 3 miles but only took Mai 24 minutes to run 3 miles. Therefore, Mai ran faster.

4. Eli thought it would be helpful to rewrite the ratios as unit rates when deciding who ran faster. His incomplete work is shown. Work with your partner to complete his work.

Mai's Workout	Jada's Workout
÷3 24 min : 3 mi ÷3 8 min : 1 mi ÷8 1 min : $\frac{1}{8}$ mi ÷8	÷3 30 min : 3 mi ÷3 10 min : 1 mi ÷10 1 min : $\frac{1}{10}$ mi ÷10

5. Complete the statements.

It took Mai <u>8</u> minutes to run 1 mile, and it took Jada <u>10</u> minutes to run 1 mile. This shows that [○ Mai ○ Jada] ran faster because it took her [○ less ○ more] time to run the same distance.

In 1 minute, Mai ran $\frac{1}{8}$ mile and Jada ran $\frac{1}{10}$ mile. This shows that [○ Mai ○ Jada] ran faster because she ran [○ less ○ more] distance in the same amount of time.

Students may not understand that a faster time means Mai ran at a faster rate, instead thinking that running for 30 minutes means Jada ran faster or more miles because 30 is more than 24.
Make sure to address this misconception if you see it arising.

- *Who ran for longer? How do you know?*
- *Did both students run the same number of miles?*
- *What does it mean if Mai ran the same number of miles in less time?*
- *Is she running faster or slower than Jada? Is $\frac{1}{3}$ more or less than $\frac{1}{10}$?*

The table provides a visual entry point. Students can use different colors to keep track of their work and see who ran faster. Encourage partners to discuss their work in the table to provide an auditory entry point.

5.5.4 Try It: Rates and Ratios in Context of Speed

Component Overview: *Students work with a partner to determine whether the slug or snail is moving faster based on the ratio of centimeters traveled to seconds. Students evaluate a statement and explain if they agree or disagree based on the ratios.*

5.5.4 Try It: Rates and Ratios in Context of Speed

1. A slug travels 3 centimeters in 3 seconds. A snail travels 6 centimeters in 6 seconds. Ethan says, "The snail was traveling faster because it went a greater distance."

   a. Discuss Ethan's statement with your partner.
   Answers will vary. Some may agree or disagree depending on their understanding of unit rates.

   b. Explain whether you agree with Ethan using ratios or rates.
   I disagree with Ethan. The slug's unit rate is $\frac{3}{3} = 1$ cm per second. The snail's unit rate is $\frac{6}{6} = 1$ cm per second. They both have the same unit rate.

Encourage students to depict the situation using a double number line, table, or other visual representation. Highlight any creative representations as a class.

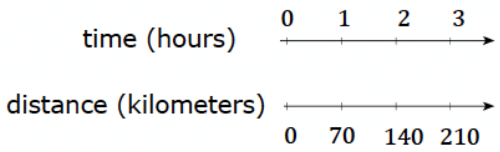
Students should be applying what they have already learned about ratios and unit rates to their discussions about Ethan's statement.
Encourage students to use the formal vocabulary of ratios, rates, and unit rates in their discourse and explanation.
Students should be evaluating the rate of the slug and snail to prove their explanations.

5.5.5 Your Turn!

Component Overview: *Students work independently to solve real-world problems involving ratios, rates, and unit rates using a double number line and a ratio table. Pose questions to check for understanding and help students consolidate their understanding after students have completed both questions.*

5.5.5 Your Turn!

1. Nadia is driving at a constant speed on a long stretch of highway on a long drive from Melbourne, Florida, to Atlanta, Georgia. Her distance traveled over time is shown on the double number line.



How far did Nadia travel at this speed for 5 hours? Explain or show your reasoning.
The unit rate of Nadia's travel is 70 kilometers per hour. Nadia traveled $70 \times 5 = 350$ kilometers.

2. Two cleaning solutions were created to clean tile floors as shown.

Mixture A	Mixture B
5 cups of water 2 cups of vinegar	15 cups of water 8 cups of vinegar

a. Explain why the mixtures will not result in the same cleaning solution.
The ratio of water to vinegar for Mixture A is $\frac{5}{2} = 2.5$. This is a $2.5{:}\,1$ ratio. The ratio of water to vinegar for Mixture B is $\frac{15}{8} = 1.875$. This is a $1.875{:}\,1$ ratio. Mixture A has a greater ratio of water to vinegar which results in a different cleaning solution.

b. Explain which solution will have a stronger smell of vinegar.
Mixture B has 1.875 cups of water per 1 cup of vinegar. Mixture A has 2.5 cups of water per 1 cup of vinegar. Therefore, Mixture B would have a stronger smell because it would be less concentrated than mixture A.

Students may struggle to explain why the mixtures will not result in the same cleaning solution. Encourage students to consider what they know about ratios and unit rates to help explain their answer.
How can you determine ratio of water to vinegar for each solution? (Compare how many cups for water and how many for vinegar)
So how could you represent this comparison visually or using math? (Double number line, table, or ratio)
Do you think the mixture with more or less vinegar will smell more strongly? Why?

5.5.6 Wrap-Up: Growth Mindset

Component Overview: *Students reflect on which topics they are still unsure about and develop a plan to improve their understanding of these topics. This activity helps students enhance their metacognitive skills and encourages them to take ownership of their learning.*

5.5.6 Wrap-Up: Growth Mindset

Complete each statement.

1. Today I am still unsure about ...

   Answers will vary. Today I am still unsure about the difference between ratios and rates.

2. To fill this gap, I intend to ...

   Answers will vary. To fill this gap, I intend to study my notes.

The 5.5.6 Wrap-Up asks students to write an explanation of which topics they are still unsure about.
Students who struggle with written expression could rate each problem from the lesson using a coding system such as:

- *0 = don't understand*
- *1 = partially understand*
- *2 = completely understand*

5.6 Using Two- and Three-Column Tables of Equivalent Ratios

Lesson Introduction

Benchmarks	MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios. MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.	Learning Targets	• I can use two- and three-column tables to display equivalent ratios. • I can solve problems using two- and three-column tables of equivalent ratios.

Benchmark Coherence	Building On	Working Toward
	MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

Essential Questions	• How can I use two- and three-column tables to display equivalent ratios? • How can I solve problems using two- and three-column tables of equivalent ratios? • When is a two- or three-column table a better representation than a double number line?
Lesson Narrative	Students explore equivalent ratios through the lens of two- and three-column tables by comparing and contrasting these models to a double number line. Students should be making connections for the application of multiple representations to display equivalent ratios and solve problems.

Component Summary	5.6.1 Warm-Up (~5 min.)	Students consider different ways of representing division *(SE p. 198)*	5.6.3 Guided Instruction (15-20 min.)	Use three-column ratio tables to display equivalent ratios and solve problems *(SE p. 200)*
	5.6.2 Exploration (15 min.)	Students explore the advantages of modeling an equivalent ratio in a two-column table over a double number line *(SE p. 199)*	5.6.4 Guided Practice (10 min.)	Practice using three-column ratio tables to display equivalent ratios and solve problems *(SE p. 202)*
	5.6.5 Wrap-Up (~5 min.)	Formative assessment opportunity using a two-column ratio table to solve real-world rate problems *(SE p. 203)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	• (Optional) Note cards • (Optional) Chart paper • (Optional) Markers	N/A

Teacher Tips

Common Misconceptions

- Students may not independently recognize or understand the advantages of using a two-column table over using a double number line.
- Students may confuse part-to-part and part-to-whole ratios.
- Students may multiply when they should be dividing or divide when they should be multiplying (i.e. students may multiply by 3 in 5.6.3 Guided Instruction question 3a rather than dividing by 3 or multiplying by $\frac{1}{3}$).

Things to Consider

- Explicitly show and explain the features of two- and three- column tables that make them more flexible than double number lines:
  - On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved.
  - A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.
- Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio. Also, have students write their calculations out so they can check their work. This will help prevent errors.

Literacy and Language Routines

Think-Pair-Share

Consider having students engage in 5.6.3 Guided Instruction using the Think-Pair-Share routine. Initially, have students independently solve the problems. Then, ask students to discuss how they solved the problems and the answer they obtained with a partner. Have students resolve any differences in their answers through discussion. The discussion will help develop student understanding and confidence.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.5.1: Patterns and Structures

The two- and three-column tables help students identify and use the equivalent ratios to solve real-world problems. Students identify the equivalent ratios in the two- and three-column tables. Then, they use the equivalent ratios to solve for unknown values. After solving a few problems, students will develop generalizations based on the similarities found among the problems.

Differentiate Instruction

Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio. For example, notice the table of values in 5.6.3 Guided Instruction. Students may apply rules of distribution, thinking similar strategies would apply. Consider proving both visual and verbal entry points for students to make sense of which values should be multiplied and the reasoning behind the operations.

Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

Lesson Notes:

5.6.1 Warm-Up: Different Ways of Expressing Division

Component Overview: *Students consider different ways division can be represented and explain connections between division and ratios.*

5.6.1 Warm-Up: Different Ways of Expressing Division

In many Latin American countries and other countries around the world, the colon (:) is used instead of ÷ to express division. Therefore, the expression "28 divided by 4" would be expressed as $28:4$ instead of $28 \div 4$.

1. Using what you have learned about ratios, explain how $28:4$ and $28 \div 4$ can be used to express the same value.

   Answers will vary.

   The ratio 7:1 is equivalent to 28:4. It can be found by dividing both 28 and 4 by 4. 28 divided by 4 is 7.

2. Write one question that comes to your mind after reading the information shown.

   Answers will vary.

   Are there any other symbols that are different in other countries that mean the same thing?

Student answers will vary, and the goal isn't for all students to get this "right." The purpose is to expose students to differences in the ways people in different parts of the world express mathematics.

After students have written their answers, consider having students Turn and Talk or share whole group other ways they have personally seen division represented, either from personal experience or from others.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

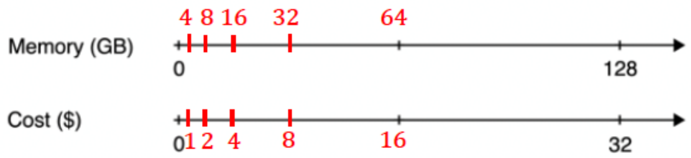
Component Overview: *Students work with a partner to explore and compare modeling equivalent ratios in two-column tables versus double number lines. Students then have a choice of which representation to use to solve the last problem, explaining their choice in a written explanation.*

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables

Work with your partner on the following.

In 2016, 128 gigabytes (GB) of portable computer memory cost \$32.

A double number line diagram that represents the situation is shown.



1. One set of tick marks has already been drawn to show the result of multiplying 128 and 32 each by $\frac{1}{2}$. Label the amount of memory and the cost for these tick marks.

2. Next, keep multiplying by $\frac{1}{2}$ and drawing and labeling new tick marks, until you can no longer clearly label each new tick mark with a number.

3. Discuss with your partner whether or not it's possible to clearly represent the cost of 1 GB of memory on the double number line diagram.

4. If it is possible, circle the cost of 1 GB of memory on the double line diagram. If it is not possible, explain why not.
   Answers will vary. One would need to circle in between 0 and 1. There isn't enough space to label this clearly on the double line diagram.

Have students intentionally compare and contrast the usefulness of tools like double number lines and tables. When would a double number line be helpful? When would a two-column table be better?
On a double number line, the differences between numbers are represented by lengths. This can be a limitation when large or small numbers are involved.
A double number line dictates the ordering of the values on the line, but in a table, pairs of values can be written in any order.

5. A table representing the same situation is shown. Work with your partner to find the cost of 1 GB. Use as many or few rows as needed.

Memory (gigabytes)	Cost (dollars)
128	32
64	16
32	8
16	4
4	1
1	0.25

Students may choose to use more rows to arrive at this answer. However, if they understand how to find the unit rate at this point (which many will), they may recognize that they can do it in one step by dividing both values by 128.

6. Explain your steps for determining the cost per gigabyte using the table.
Answers will vary. Divide 128 by 2 and until you get to 16 and then begin dividing by 4.

7. Use either the double line diagram or the ratio table to determine the cost of 512 GB of portable computer memory at this rate.

Memory (gigabytes)	Cost (Dollars)
128	32
512	128

Multiply both values by 4. The cost for 512 gigabytes is $128.

Consider allowing students to share their answers verbally prior to writing their answers. Sharing their answer verbally will allow them to rehearse their explanation and receive feedback. This will improve their confidence and the quality of their responses.

Poll the Class
Consider polling the class to see which students prefer the table and which prefer the double number line. This can lead to a discussion of the advantages of both representations, and the teacher can highlight when it makes the most sense to use each representation.

5.6.2 Exploration: Double Number Lines Versus Two-Column Tables (continued)

8. Discuss with your partner which representation of the equivalent ratios you preferred using to find the costs of 1 GB and 512 GB. Summarize your discussion.
I preferred to use a table because it helps me to visualize the number relationships.

The IBM 7094 was regarded as one of the most powerful advanced mainframe computers in the early 1960s. NASA and the Air Force used it in the Gemini and Apollo space programs.

Despite its large size and historical impact, the IBM 7094 only had a core memory of about 150 kilobytes (KB). That's only enough to manage a few Microsoft Word documents.

Today, you can get a small USB flash drive like the one shown for less than $20 that can store 4 gigabytes (GB) of memory.

That's almost 27,000 times the memory as the IBM 7094, and you can fit it in your pocket!

Share history or a clip from the movie, Hidden Figures to engage more learners in the impact of IBM and important mathematical figures on our technology today.

Mathematician Mary Jackson, pictured below, is one of the key mathematical figures featured in the movie. She and women like her performed all calculations by hand in the early pre-computer days at NASA.

5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Component Overview: *Students use a table to find and represent the number of free-throw shots Kiara took at basketball practice. Students find missing values and then explain how they used the table to do so. Students then apply this to a table representing the concrete mixture, using information in the table to find other missing information.*

5.6.3 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	6
16	8	24
2	1	3

   a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.

   b. How was each value in the second row determined?

   We are told that she missed 8 shots. Previously she missed 2 shots. Dividing 8 by 2 we obtain 4. This tells us that each value in row one should be multiplied by 4. Multiplying 4 by 4 we see she made 16 shots. Multiplying 6 by 4 we see she took a total of 24 shots. Checking our work, the shots made and shots missed equals the total shots.

Notice the opportunities for verbal mathematical reasoning in this component. Some students may benefit from verbalizing their responses (questions 1b and 2a) before putting them in writing. Ensure that students have an opportunity to engage in discussion to provide this entry point for learners.

Think-Pair-Share
To provide an opportunity for student discussion, engage the class in a "Think-Pair-Share" routine for the first question. Initially, have students independently solve the problems. Then, have students discuss how they solved the problems and the answers they obtained with a partner. Have students resolve any differences in their answers through discussion. As students are discussing, circulate to collect informal data on how students are determining the values in each row. Consider bringing the class back as a group and having students share particularly efficient strategies or draw attention to a common misconception.

5.6.3 Guided Instruction: Using Three-Column Ratio Tables (continued)

2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.

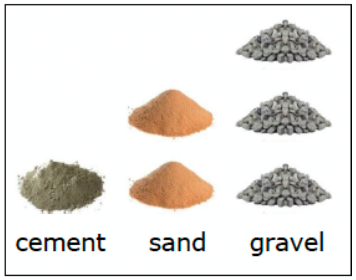
   a. How was each value in the third row determined?
   We are told that she took 3 shots. Previously she took 6 shots. Dividing 6 by 3 we obtain 2. This tells us that each value in row one should be divided by 2. Dividing 4 by 2 we see she made 2 shots. Dividing 2 by 2 we see she missed a total of 1 shot. Checking our work, the shots made and shots missed equals the total shots.

   b. Complete the statement.

   For every 3 free-throw shots taken, Kiara makes _2_ shot(s) and misses _1_ shot(s) on average. The ratio of missed shots to made shots to total shots is 1 : 2 : 3.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

   Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



To encourage mathematical reasoning as students fill in the tables, ask questions like:

- *Do the answers in your tables make sense? How do you know?*
- *How did you use the table to help determine what should go in the next row?*
- *How can you check your work to ensure you filled in the table correctly?*

a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Divide by 3
Multiply by 6.5
Multiply by 5

Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?
Marc will need 6 gallons of dry mix to make one batch of concrete.

Challenge advanced learners to write their own real-world problems that can be displayed and solved using a three-column ratio table. Students can solve their own problems or swap with a partner and solve each other's problems.

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

Component Overview: *Students begin this Guided Practice by using a three-column table to find the quantities needed for different paint mixtures, then explain how they completed the table. Students then complete two more tables based on given real world contexts, applying their knowledge of ratios and unit rates to find unknowns in the tables.*

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	8
20	12	32
1.25	0.75	2
2.5	1.5	4
50	30	80

   a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?
   Ten batches are in 80 cups of maroon paint.

   b. Explain how this relates to the factor used to complete the last row in the table.
   Each number of cups in the first row was multiplied by 10 to determine the number of cups in the last row. Therefore, if the first row represents one batch, the last row will represent 10 batches.

To provide a visual entry point, have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio.

5.6.4 Guided Practice: Using Two- and Three-Column Ratio Tables (continued)

2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons
16	1
80	5

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	24
Non-Walking Days	1	6
Total Days	5	30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked 24 of the 30 days.

After students complete the tables in questions 1, 2 and 3 consider having students share their strategies in small groups. This will provide an opportunity for students to verbalize their thinking and compare their strategies to others' means of solving, allowing them to make connections and/or determine more efficient means of solving.

5.6.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their understanding of using two-column ratio tables to solve real-world rate problems by determining the distance traveled when walking at a given rate.*

5.6.5 Wrap-Up: Ticket Out the Door

1. Julio participated in a walk-a-thon to raise money for cancer research. Walking at a constant speed, Julio travelled 6.4 miles in the first 2 hours of the walk-a-thon. His goal is to walk for 7 hours. Use the table to find how far Julio will have walked if he reaches his 7-hour goal. Show how the values in each row are determined.

Hours	Miles Walked
2	6.4
1	3.2
7	22.4

$\frac{1}{2}\cdot$ $\cdot\frac{1}{2}$

$7\cdot$ $\cdot 7$

Julio will walk 22.4 miles in 7 hours.

Consider using EdgeXL to provide additional enrichment and practice opportunities for students.

5.7 Digging Deeper Into Unit Rates

Lesson Introduction

Benchmarks	MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate. MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.	Learning Targets	• I can determine a unit rate when given different ratios. • I can interpret unit rates in real-world contexts. • I can use unit rates to solve problems.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

Essential Questions	• How do you determine a unit rate when given different ratios? • How do you interpret unit rates in real-world contexts? • How do you use unit rates to solve problems?
Lesson Narrative	Students continue exploring unit rates in a variety of contexts, including carnival tickets (5.7.1 Warm-Up), working and job pay (5.7.2 Guided Practice), and household purchases (5.7.4 Collaboration). Opportunity for differentiation in both content and process exist throughout the lesson, with students using tape diagrams explicitly during 5.7.2 Guided Instruction before utilizing strategies of three-column methods beyond. 5.7.4 Collaboration and 5.7.5 Bring It Together provide a variety of entry points for application and synthesis before formative assessment in 5.7.6 Wrap-Up.

Component Summary				
	5.7.1 Warm-Up (~10 min.)	Compare the unit rates of carnival tickets *(SE p. 204)*	5.7.4 Collaboration (10-15 min.)	Collaboratively interpret and use unit rates to solve real-world problems *(SE p. 207)*
	5.7.2 Guided Instruction (10-20 min.)	Use tape diagrams to interpret unit rates *(SE p. 204)*	5.7.5 Bring It Together (5 min.)	Write three statements to summarize learning about unit rates *(SE p. 208)*
	5.7.3 Guided Practice (10 min.)	Practice using unit rates to solve real-world problems *(SE p. 206)*	5.7.6 Wrap-Up (~5 min.)	Formative assessment opportunity using unit rates for real-world problems *(SE p. 208)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: N/A	N/A	N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may automatically think the larger quantity should be first when determining the unit rate.
Students may multiply when they should be dividing or divide when they should be multiplying.
Students may make computational errors if they have not mastered their multiplication and division of rational numbers.</td><td>Consider allowing students to use a calculator.
Remind students that unit rates will always be rounded to the nearest hundredth when dealing with money.
Making a no-bake recipe in class would provide a kinesthetic entry point.
Comparing unit rates of products using online or newspaper flyers would reinforce the utility of finding the best value.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Consider having students engage in 5.7.1 Warm-Up using the Think-Pair-Share routine. Initially, have students independently solve the problems. Then, have students discuss how they solved the problems and the answers they obtained with a partner. Have students resolve any differences in their answers through discussion. The discussion will help develop student understanding and confidence.</td><td>MA.K12.MTR.7.1: Real-World Contexts
This lesson presents a multitude of opportunities for rich dialogue on real-world applications. The applications in this lesson include carnival ticket prices (5.7.1 Warm-Up), hiking and biking rates (5.7.2 Guided Instruction), rate of pay (5.7.3 Guided Practice), unit prices (5.7.4 Collaboration), and ratios of ingredients in a cheesecake recipe (5.7.6 Wrap-Up). Students use ratio tables and tape diagrams to display equivalent ratios and solve problems in these contexts.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider placing students in groups and assigning groups to each number of the 5.7.4 Collaboration. By differentiating which groups engage with each portion of the component, the lesson can become a jigsaw in which students are responsible for sharing their learning with others.
Jigsaw increases student agency by offering choice. Additionally, students are positioned as experts on their particular question, centering their approach and experience as expertise. Finally, the teacher could prompt students to share their partner’s work rather than their own, further positioning students as analytical thinkers in processing and communicating the work of others.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.7.1 Warm-Up: Ticket Booth

Component Overview: *Students are shown a poster of the price of different numbers of tickets at a festival and make predictions about which option is the best. Students then determine if there are any equivalent ratios on the poster and then find the factor by which the ratios need to be multiplied to find the second ratio.*

5.7.1 Warm-Up: Ticket Booth

At a local fall festival, the following prices are shown at the ticket booth.

Ticket Prices	
1 Ticket	$0.50
12 Tickets	$5.00
25 Tickets	$10.00
50 Tickets	$25.00
120 Tickets	$50.00

1. Without performing any calculations, explain which option you think would be the best.

   Answers will vary.

   It depends on how many tickets you want. The unit rates are: 50 cents (if you buy 1 ticket), $0.41\overline{6}$ cents (if you buy 12 tickets), 40 cents (if you buy 25 tickets), 50 cents (if you buy 50 tickets), $0.41\overline{6}$ cents (if you buy 120 tickets). The best value is 25 tickets.

2. Determine a pair of ratios, tickets:price, that appear to be equivalent. Write them in the space below.

   If it is not possible to determine a pair of equivalent ratios from the sign, select one ratio and write another one that is equivalent.
   Either of the following are correct:
   _1_ ticket(s) : $_0.50_

   _50_ ticket(s) : $_25_

   These are both 50 cents per ticket.

   _12_ ticket(s) : $_5_

   _120_ ticket(s) : $_50_

   These are both $0.41\overline{6}$ per ticket.

Ticket Booth – Token Economy
Consider utilizing a token economy for encouraging strong Mathematical Thinking and Reasoning (MTR) Standards throughout classroom discussion. When students analyze the thinking of others (MA.K12.MTR.4.1), make connections using patterns or structure (MA.K12.MTR.5.1), or utilize multiple representations when problem solving (MA.K12.MTR.2.1), the tickets can serve as positive reinforcement while also acting as an opportunity to apply learning of ratios when selecting prizes – like this Ticket Booth!

5.7.1 Warm-Up: Ticket Booth (continued)

3. By what factor can both values in the first ratio be multiplied to find the second ratio?

   _1_ ticket(s) : $_0.50_

   _50_ ticket(s) : $_25_

   Multiply the number of tickets and cost both by 50.

   _12_ ticket(s) : $_5_

   _120_ ticket(s) : $_50_

   Multiply the number of tickets and cost both by 10.

5.7.2 Guided Instruction: Interpreting Unit Rate

Component Overview: *In this Guided Instruction, students review the definition of unit rate and apply this to identifying and finding the unit rates for several scenarios. Students then use tape diagrams to represent and interpret unit rates for different speeds, identifying which person is moving slower based on their unit rates.*

5.7.2 Guided Instruction: Interpreting Unit Rate

A ***unit rate*** is a ratio comparing a number of units of a quantity to 1 unit of another quantity.

1. Several rates are listed. Circle those that represent a unit rate.

earning $42 for 3 hours

35 miles per gallon

75 miles per hour

2.5 miles in 24 minutes

each banana costs $0.29

12 inches for every 1 foot

2. Tia was hiking on the Florida National Scenic Trail. She hiked a total of 4.5 miles in 2 hours. The tape diagram models the distance and time she hiked.

Tia hiked 4.5 miles.	*2.25 miles*	*2.25 miles*
Tia hiked for 2 hours.	*1 hour*	*1 hour*

Notice that each tape diagram is the same size because they represent quantities in the same situation.

a. Fill in each box of the tape diagram to show the value it represents.
See tape diagram.

b. What is Tia's unit rate in miles per hour?
2.25 miles per hour

This component is intentionally designed to engage various learning styles. The tape diagrams provide a visual entry point for students and allow them to visualize and interpret different unit rates. Allow students to engage in verbal discourse to provide an auditory entry point.

5.7.2 Guided Instruction: Interpreting Unit Rate (continued)

c. Draw a tape diagram that represents the time it would take Tia to hike 9 miles on the trail at this pace.

Tia hiked 9 miles.

2.25 mi.	2.25 mi.	2.25 mi.	2.25 mi.

Tia hiked for 4 hours.

1 hr.	1 hr.	1 hr.	1 hr.

3. Carter bikes $12\frac{1}{2}$ miles in 5 hours around the city.

a. Work with your partner to fill in each box of the tape diagram to show the value it represents. Be sure to include units.

2.5 miles	2.5 miles	2.5 miles	2.5 miles	2.5 miles

1 hour	1 hour	1 hour	1 hour	1 hour

b. Interpret the unit rate for Carter's bike ride. Include units.

2.5 miles per hour

c. Explain which person is moving slower, Tia or Carter. Include the unit rates in your explanation.

Tia is moving slower, because she travels a shorter distance (2.25 miles versus 2.5 miles) in the same amount of time (1 hour).

4. Complete the statement.

In each scenario, the unit rate in ○ hours per mile ○ miles per hour can be determined by ○ dividing ○ multiplying the number of ○ hours ○ miles traveled by the number of ○ hours ○ miles traveled.

This Guided Instruction is intentionally positioned to guide students through the representation of unit rates using tape diagrams. Allow students to work with a partner to create each tape diagram and circulate the room to observe how students are representing the unit rates. Review the representations as a class before releasing students to finish this component.

Consider using talk moves like "revoice" or "reword" to give students an opportunity to synthesize their findings in their own words.
"In your own words, explain how you might use a tape diagram to determine a unit rate."
"In your own words, how might you find the unit rate in a problem-solving situation?"

5.7.3 Guided Practice: Using Unit Rates to Solve Problems

Component Overview: *Students revisit the ticket booth problem from 5.7.1 Warm-Up and use the unit rate of cost per ticket to complete a three-column table. Students then work with a partner to determine which ticket option is the best and which are equivalent and compare their answers to the Warm-Up. Partner pairs then use a table to answer more questions about the hourly rate of three employees.*

5.7.3 Guided Practice: Using Unit Rates to Solve Problems

Unit rates would be helpful in considering the ticket booth from the Warm-Up. Miguel set up the table shown to consider the costs per ticket for each advertised price.

1. Work with your partner to complete the table. Then, answer the questions that follow.

Total Cost	Number of Tickets	Cost per Ticket
\$0.50	1	\$0.50
\$5.00	12	$\$0.41\overline{6}$
\$10.00	25	\$0.40
\$25.00	50	\$0.50
\$50.00	120	$\$0.41\overline{6}$

a. Using the unit rates, explain which ticket option(s) is (are) the best deal.
The best deal is the 25 tickets for \$10.

b. Using the unit rates, explain which ticket options are the same deal.
The 1 ticket for \$0.50 and the 50 tickets per \$25.00 are both \$0.50 per ticket.

The 12 ticket for \$5.00 and the 120 tickets per \$50.00 are both $\$0.41\overline{6}$ per ticket.

c. Compare your answer to the previous question to the equivalent ratios you determined in the Warm-Up. Make any revisions to your previous thinking, as necessary.

d. With your partner, discuss recommendations for how the high school should change its prices for carnival tickets.

Notice the partner work in this component. Be intentional with grouping, possibly using data or informal observations from the Lesson 6 Wrap-Up to inform pairing.

Scaffold this component with intentional questions like:
- *Was it a better value to buy more or fewer tickets? How do you know?*
- *Based on the ticket prices and unit rate, how can the high school change its prices for carnival tickets? Why would this be helpful?*

Prompt students toward discussing unit price and its role in determining lower prices in bulk. Normally larger amounts are priced cheaper per unit – just like the cost of most items at the store is cheaper per unit the more you buy, so is the cost per ticket.

5.7.3 Guided Practice: Using Unit Rates to Solve Problems (continued)

2. Holly, Frank, and Ella work part-time for Shoes-R-Us. The table shows how much money each person earns during a work week.

Employee	Weekly Pay	Number of Hours Worked
Holly	$165	20
Frank	$162	18
Ella	$170	20

Work with your partner to answer the following questions using the information in the table.

a. Find and interpret the hourly rate Holly earns.
Holly earns $8.25 per hour. This means that each hour she earns $8.25.

b. Which employee earns the higher hourly rate, Ella or Frank?
Frank earns $9.00 per hour. Ella earns $8.50 per hour. Therefore, Frank earns the higher hourly rate.

c. If Frank works 25 hours next week, how much should he expect to earn?
Frank will earn $225.00 after working 25 hours.

Encourage student discourse throughout this component as they work in partner pairs.
Highlight and surface whole group any student responses that connect hourly rate as equivalent to unit rate because it is per 1 hour.

5.7.4 Collaboration: Using and Interpreting Unit Rates

Component Overview: *Students collaboratively interpret and use unit rates to solve real-world problems.*

5.7.4 Collaboration: Using and Interpreting Unit Rates

Work with your partner to interpret the unit rates for each situation.

1. Tyler paid $\$12$ for 4 large candy bars.
   a. Tyler interprets the unit rate as $\frac{\$3}{1\text{ bar}}$.
      His friend Ben interprets the unit rate as $\frac{1\text{ bar}}{\$3}$. Which person interpreted the unit rate correctly?
      Both Tyler and Ben are correct.
   b. Based on the unit rate, how many candy bars can Tyler purchase for $\$45$?
      Tyler can purchase 15 candy bars for $\$45.00$.
2. Aaliyah's family drinks a total of 10 gallons of milk every 6 weeks and 4 gallons of apple juice every 2 weeks.
   a. Which beverage does Aaliyah's family drink more of in one week?
      Aaliyah's family drinks more apple juice (2 gallons) than milk ($1\frac{2}{3}$ gallons) in one week.
   b. How many weeks does it take the family to consume 1 gallon of milk?
      It takes the family $\frac{6}{10}=\frac{3}{5}$ of a week (or 4.2 days) to consume 1 gallon of milk.
3. Carlos buys 24 bottles of spring water for $\$3.60$ and 3 jars of pasta sauce for $\$9.69$.
   a. What does each item, a bottle of spring water and a jar of pasta sauce, cost per unit?
      The spring water costs $\$0.15$ per bottle. The pasta sauce costs $\$3.23$ per jar.
   b. What would be Carlos' total cost if he purchased 10 bottles of spring water and 2 jars of pasta sauce at their unit prices? *Ten bottles of spring water would cost $\$1.50$. Two jars of past sauce would cost $\$6.46$. The total cost would be $\$7.96$.*

Consider positioning students to discuss question prompts rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation such as "To find the unit rate, I…"

Consider using a jigsaw approach for engaging students.

- *Question 1: Lower entry point with one-step multiplication and division*
- *Question 2: Accessible on grade level involving operations with fractions*
- *Question 3: Higher entry point for students who should engage with decimal operations*

Students should then find a partner with a different number and explain both their process and product. The teacher should then call on students to explain the work of their partner, further extending the mathematical thinking of self and others.

5.7.5 Bring It Together: Interpreting Unit Rates

Component Overview: *Students work with a partner to consolidate their learning by writing three statements to summarize learning about unit rates. Consider reviewing 2 or 3 statements with the class before moving on if time permits.*

5.7.5 Bring It Together: Interpreting Unit Rates

Work with your partner to write three statements to summarize your learning about unit rates from today's lesson. *Answers may vary.*

1. *I learned that to find a unit rate, you divide both quantities by the value of one of the quantities.*

2. *I learned that for unit rates expressing cost per unit, the smallest unit rate is considered the best value.*

3. *I learned that the unit rate can be used to determine other equivalent ratios.*

Consider sentence stems if you are targeting specific learning goals or consider having the students answer each of the three Essential Questions as their statements.

5.7.6 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their understanding of interpreting and using unit rates to solve real-world problems.*

5.7.6 Wrap-Up: Ticket Out the Door

A cheesecake recipe says, "Mix 12 ounces (oz) of cream cheese with 15 oz of sugar."

1. How many ounces of cream cheese are needed per ounce of sugar?
   According to the recipe, $\frac{12}{15} = \frac{4}{5} = 0.8$ *ounces of cream cheese are needed per ounce of sugar.*

2. How many ounces of sugar are used for each ounce of cream cheese?
   According to the recipe, $\frac{15}{12} = \frac{5}{4} = 1.25$ *ounces of sugar are needed per ounce of cream cheese.*

3. Max plans to make more than one cheesecake. How much cream cheese will she need to mix with 35 oz of sugar?
   Max will need 28 ounces of cream cheese to mix with 35 ounces of sugar.

Prompt students to create an additional representation of any of the questions in 5.7.6 Wrap-Up. Their choice of double number line or table will provide additional points of data when creating small groups in the next lesson.

5.8 Applying Ratio Relationships to Percentages

Lesson Introduction

Benchmark	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.	Learning Targets	• I can make connections between ratios and percentages. • I can use double line diagrams, tape diagrams, and ratio tables to solve problems involving

Benchmark Coherence	Building On	Working Toward
	N/A	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

Essential Questions	• How would you describe a percentage in terms of a ratio? • How can you use double number lines, tape diagrams, and ratio tables to solve percent problems?
Lesson Narrative	Students continue applying models of double number lines, tape diagrams, and ratio tables specifically in contexts of percentages and problems with percentages in the real-world. Students make sense of what percentage means through the lens of a ratio in order to approach problem-solving situations involving percentages similarly to previous lessons using a variety of models and strategies.

Component Summary	5.8.1 Warm-Up (~5 min.)	Estimate the percentage – orienting students to content *(SE p. 209)*	5.8.4 Guided Practice (10 min.)	Jigsaw practice for a variety of models *(SE p. 212)*
	5.8.2 Refresher (10 min.)	Review the concept of percent by representing percentages with a variety of models *(SE p. 210)*	5.8.5 Your Turn! (5-10 min.)	Demonstrate understanding of using double number lines, ratio tables, and tape diagrams to solve percent problems *(SE p. 213)*
	5.8.3 Guided Instruction (15-20 min.)	Using a variety of models for problem-solving with percentages *(SE p. 211)*	5.8.6 Wrap-Up (~5 min.)	Students reflect on their mastery of the lesson's learning targets *(SE p. 213)*

Resources	Glossary	Materials	Additional Preparation
	<u>Key Term</u>: • Percent <u>Additional Terms</u>: N/A	N/A	N/A

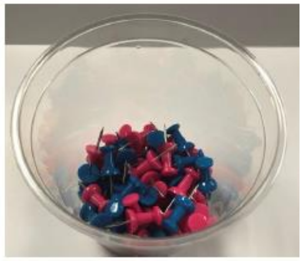
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not make the same number of partitions when creating double number lines and tape diagrams (incorrect labeling or scaling).
Students may not recognize that part-to-whole ratios connect to percentages rather than part-to-part ratios.</td><td>As an additional support, have students check that they have the same number of partitions in their tape diagram and double number lines.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
Consider having students engage in the 5.8.4 Guided Practice using the Think-Pair-Share routine. Students should independently solve the problems before discussing how they solved the problems and the answer they obtained with a partner. Encourage students to talk about their process more than their product, making note and surfacing any difference and variety in problem-solving techniques.</td><td>MA.K12.MTR.2.1: Multiple Representations
Throughout the lesson, students see ratios and percentages represented in multiple ways. Facilitate discussion to highlight a variety of strategies and representations, particularly noting differences in approach and strategy as students complete 5.8.4 Guided Practice.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">5.8.2 Refresher provides a visual entry-point for students to visualize percentages. Consider providing students with graph paper and colored pencils so they can color a visual model of the percentages.
Consider allowing students to discuss their solution strategies and answers verbally with a partner for the 5.8.4 Guided Practice and 5.8.5 Your Turn! components. Discussing the problems will help develop student understanding and confidence.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.8.1 Warm-Up: Estimate the Percentage

Component Overview: *Students estimate the percentage of straws that are orange from an image of a bunch of orange and black straws.*

5.8.1 Warm-Up: Estimate the Percentage

A photo of blue and pink thumb tacks is shown.



1. Based on what is shown in the picture, predict the percentage of thumb tacks that are blue by completing each statement. *Answers will vary.*

   I know for sure that more than __30__% of the thumb tacks are blue.

   I know for sure that less than __90__% of the thumb tacks are blue.

2. My best estimate is that __45__% of the thumb tacks are blue.

Turn and Talk
Consider having students participate in a "Turn and Talk" routine before answering question 2.
How can you use your estimates from question 1 to help you estimate for question 2?

5.8.2 Refresher: Percentages

Component Overview: *Students review the concept of percent by representing percentages with area models, number lines, fractions, and percentages. Read the meaning of percent out loud for the class, then have students work in partners to represent percentages in different ways.*

5.8.2 Refresher: Percentages

A ***percentage*** is a ratio expressed as a fraction of 100.

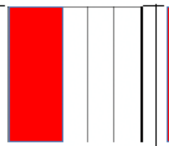
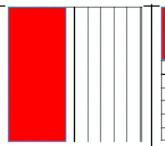
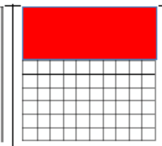
> ***Percent*** (%) means "per 100," "for every 100," or "out of 100."

Work with your partner to complete the following.

1. Each of the large squares shown are the same size.

   a. Shade $\frac{2}{5}$ of each model.

   b. Then, express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction	$\frac{2}{5}$	$\frac{4}{10}$	$\frac{40}{100}$
Percentage	$\frac{2}{5} \times 100 = 40\%$	$\frac{4}{10} \times 100 = 40\%$	$\frac{40}{100} \times 100 = 40\%$

Consider providing students with graph paper and colored pencils so they can color a visual model of the percentages or encouraging students to model percentages by folding a piece of paper to provide an entry point for kinesthetic learners.

5.8.2 Refresher: Percentages (continued)

c. Label the identified tick marks on each number line below to match the models.

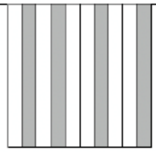
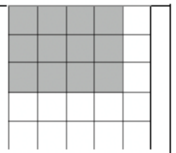
0 — $\frac{2}{5}$ — $\frac{5}{5}$

0 — $\frac{4}{10}$ — $\frac{10}{10}$

0 — $\frac{40}{100}$ — $\frac{100}{100}$

0% — 40% — 100%

2. Express the shaded part of each model as a fraction and as a percentage.

Model			
Fraction	$\frac{4}{5}$	$\frac{4}{10}$	$\frac{12}{25}$
Percentage	$\frac{4}{5} \times 100 = 80\%$	$\frac{4}{10} \times 100 = 40\%$	$\frac{12}{25} \times 100 = 48\%$

3. Complete the statements.
Percentages represent ○ part-to-part ○ part-to-whole relationships where each percentage is a part per 100. In this case, 100% is ○ only part of the total. ○ the total, or whole, amount.

Compare and Connect
After students have completed the table, consider engaging students in a "Compare and Connect" routine to establish the connections between the multiple ways to represent percentages. To facilitate conceptual understanding ask questions like:

- *How does the model represent $\frac{2}{5}$?*
- *Why do you need to multiply $\frac{2}{5}$ by 100 to find the percent?*
- *What do you notice about all of the models and the fractions they represent?*

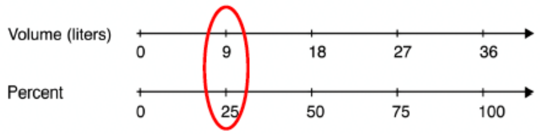
5.8.3 Guided Instruction: Using Ratio Models to Represent Percentages

Component Overview: *This Guided Instruction begins with students using a double number line representation to find the percentage of water in a pot at different intervals. Students then connect the double number line to a table representation and explain how to they relate to one another. Students then explore how to represent percentages using tape diagrams and use tape diagrams to determine the number of students in each grade based on their percentage on the tape diagram.*

5.8.3 Guided Instruction: Using Ratio Models to Represent Percentages

Double number lines, tape diagrams, and ratio tables can be used to model percentages similarly to other ratios. These can be helpful for determining equivalent ratios and solving problems involving percentages.

1. Consider a cooking pot that can hold 36 liters of water. The double number line diagram models this situation.



a. What percentage of the pot is filled when it contains 9 liters of water?
25%

b. Circle where the answer to the previous question is represented on the number line diagram.

c. How many parts were the totals on each number line, 36 liters and 100%, split into?
4 parts

d. An equivalent ratio table can also be used to model similar thinking. Determine the factor that was multiplied by 36 to get 9, and use it to complete the blanks below.

Volume (liters)	Percentage
36	100
9	25

$\times \frac{1}{4}$ $\times \frac{1}{4}$

e. Describe how the factor used to find the equivalent ratio in the table relates to how the number lines were divided in the number line diagram.
The factor used is to find the equivalent ratio is $\frac{1}{4}$. This relates to how the number lines were divided because each of them was divided into fourths.

MA.K12.MTR.2.1: Multiple Representations
In this component, students see ratios and percentages represented in multiple ways. They are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout this component. These representations help students develop a conceptual understanding of percentages and how they are related to rates and ratios. Consider incorporating questions that position students to evaluate which representation makes the most sense to them in each unique context.

5.8.3 Guided Instruction: Using Ratio Models to Represent Percentages (continued)

Tape diagrams can also be used to represent problems involving percentages.

2. At Millennium Middle School, there are 70 students in the school band. Of the band members, 40% are sixth graders, 20% are seventh graders, and the rest are eighth graders.
   The tape diagrams shown represent the situation.

band members	14	14	14	14	14
percentage	20%	20%	20%	20%	20%

   a. Label each small rectangle in the tape diagram of the band members with the number of students it represents.

   b. Label each small rectangle in the tape diagram of the percentage of band members with the value it represents.

   c. How many band members are sixth graders?
   28 *band members are* 6th *graders.*

   d. Circle the part of the tape diagram that represents these students.

   e. How many band members are seventh graders?
   14 *band members are* 7th *graders.*

   f. Draw a rectangle around the part of the diagram that represents these students.

   g. What percentage of the band members are eighth graders?
   40% *are* 8th *graders.*

   h. How many band members are eighth graders?
   28 *band members are* 8th *graders.*

This Guided Instruction is intentionally scaffolded so students can draw connections between multiple visual representations and percentages.
Guide students through question 2a and 2b as they determine how to use the tape diagrams to represent percentages of each grade in the school band.
Then, have students work in partners or small groups to use their tape diagrams to answer the questions. Circulate to observe how students are using their representations to make sense of each percentage.

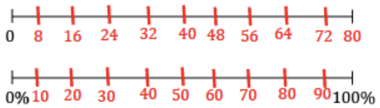
Students might have trouble determining the number of students in each grade, especially because the total number of students is mentioned at the beginning of this component and not connected directly to the tape diagram.
Encourage students to label their tape diagrams to show the total number of students and ask probing questions to guide them through the process of determining the students in each grade.
Which section of the tape diagram represents the number of students in $6^{th}/7^{th}/8^{th}$ grade? How can you use this to find the total?
How can you determine the number of students represented in each section of the tape diagram? Label each section if it will help keep track.

5.8.4 Guided Practice: Using Ratio Models to Represent Percentages

Component Overview: *Students practice using double number lines, ratio tables, and tape diagrams to solve percent problems.*

5.8.4 Guided Practice: Using Ratio Models to Represent Percentages

1. Use the number line diagram to determine what percentage of 80 is 24. Then, complete the statement.



24 is 30% of 80.

2. Use the tape diagram to determine 70% of \$60. Then, complete the statement.

dollars	6	6	6	6	6	6	6	6	6	6
percentage	10	10	10	10	10	10	10	10	10	10

70% of \$60 is \$_42_.

3. During a sale, a pair of jeans is on sale for 80% of its regular price. The regular price of the jeans is \$75. Complete the ratio table to determine the sale price. Then, complete the statement.

Price (\$)	Percentage
\$60	80
\$75	100

80% of \$75 is \$_60_.

MA.K12.MTR.2.1: Multiple Representations
Notice the varying representations found in questions 1-3. When students have completed all three questions, consider prompting students to make connections between those representations and their effectiveness in problem-solving.
Why was a number line used in question 1? Would a tape diagram or a table have been more effective? Why or why not? (answers will vary, encourage creativity and student choice)
What about the tape diagram in question 2 or the table in question 3? Did those representations make the most sense in those contexts? Did you use something else? Why or why not?

5.8.5 Your Turn!

Component Overview: *Students work independently to demonstrate their understanding of using double number lines, ratio tables, and tape diagrams to solve percent problems.*

5.8.5 Your Turn!

1. Use the models to find the values that complete each statement. Show your work.
   For the last row, use the model of your choice.

<table>
<tr><th>Statement</th><th>Model</th></tr>
<tr><td>20% of 60 is 12.</td><td>0   12   24   36   48   60
0%   20%   40%   60%   80%   100%</td></tr>
<tr><td>48% of 25 is 12.</td><td><table><tr><th>Value</th><th>Percentage</th></tr><tr><td>25</td><td>100</td></tr><tr><td>12</td><td>48</td></tr></table></td></tr>
<tr><td>40% of 70 is 28.</td><td><table><tr><td>value</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td><td>7</td></tr><tr><td>%</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td><td>10</td></tr></table></td></tr>
<tr><td>16% of 125 is 20.</td><td><table><tr><th>Value</th><th>Percentage</th></tr><tr><td>125</td><td>100</td></tr><tr><td>20</td><td>16</td></tr></table>
Answers will vary.</td></tr>
</table>

This component allows students to use the model of their choice to determine the last row in the table, allowing for multiple entry points to demonstrate understanding and learning.
Consider how student responses in this component could inform future small-group planning.

5.8.6 Wrap-Up: Reflection

Component Overview: *Students reflect on their mastery of the lesson's learning targets.*

5.8.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can make connections between ratios and percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary.

2. I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

Answers will vary.

3. Write one question you still have about solving problems involving percentages after today's lesson.
   Answers will vary.

Consider using the following prompts as an additional data point for more accurate reflection:

- *Which questions from the lesson involved making connections between ratios and percentages?*
- *Code them and think about how confident you felt answering these questions.*
- *Which questions from the lesson involved using double number lines, tape diagrams, and ratio tables to solve problems involving percentages?*
- *Code them and think about how confident you felt answering these questions.*

Coding can include Mark-Up Texts like √ for confidence, ? for questions or help needed, or ± for getting there but need more practice.

5.9 Applying Ratios to Solve Problems Involving Percentages

Lesson Introduction

Benchmark	MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.	Learning Target	• I can apply ratio relationships to solve problems involving percentages.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

Essential Questions	• How do you apply ratio relationships to solve problems involving percentages? • In a real-world context, how do you identify the value for the part? • In a real-world context, how do you identify the value for the whole?
Lesson Narrative	In this lesson, students extend their understanding of ratios and percentages to apply them to real-world contexts. Students begin by using double number lines, tape diagrams, and tables to find equivalent ratios and their corresponding percentages. Students continue to build comprehension through using the more abstract method of representation, tables, to find missing values and percentages based on a context and ratio. The lesson culminates in students demonstrating their ability to use equivalent ratio tables to calculate percentages in real world contexts.

Component Summary	5.9.1 Warm-Up (~5 min.)	Use a double number line to estimate percentages *(SE p. 214)*	5.9.3 Exploration (15-20 min.)	Explore using equivalent ratio tables to calculate percentages in real-world contexts *(SE p. 216)*
	5.9.2 Guided Practice (5-10 min.)	Apply ratio relationships to solve real-world percentage problems *(SE p. 215)*	5.9.4 Your Turn! (10 min.)	Practice using equivalent ratio tables to calculate percentages in real-world contexts *(SE p. 217)*
	5.9.5 Wrap-Up (~5 min.)	Students demonstrate the use of two-column tables to determine ratios and percentages of a real-world situation *(SE p. 218)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms</u>: N/A	N/A	N/A

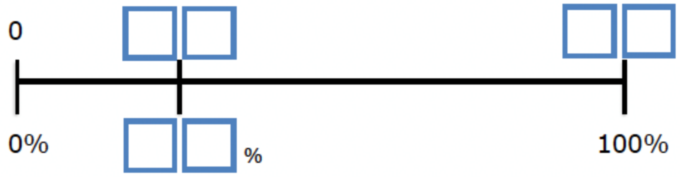
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may accidentally multiply on one side and divide on the other side of the equivalent ratio table, as they may do in solving equations.
Students may think that the equivalent ratio in tables must be a whole number (see “fractions of a person” question in 5.9.3 Exploration).</td><td>Utilize data or informal observations from Lesson 8 to drive decision-making in instruction and practice for this lesson, as this lesson is primarily building on the knowledge on Lesson 8 in practice for Lesson 10.
Look at Lesson 11 when planning for this lesson to determine where your instruction and student practice can be spiraled.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
Consider pairing 5.9.3 Exploration with a Notice and Wonder routine. This will position students to make connections between ratios and percentages to navigate a variety of real-world contexts using different methods, particularly the methods and patterns that make the most sense to them as learners.</td><td>MA.K12.MTR.2.1: Multiple Representations
Students apply ratios relationships to solve real-world problems involving percentages. Across the lesson components, the ratios are represented in seven equivalent ratio tables, two double number lines, and one tape diagram. Making connections among these representations deepens student understanding.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Consider using the double number line or equivalent ratio table to help students visualize how to write the ratio as a percentage in all components of the lesson. If one representation is provided in the lesson, consider also displaying the problem using the other representation.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.9.1 Warm-Up: Percentages on the Number Line

Component Overview: *Students use a double number line to estimate percentages.*

5.9.1 Warm-Up: Percentages on the Number Line

1. Place a digit in each blue box to create an accurate number line. Use only digits 0 to 9, at most one time each. The number line is not drawn to scale.



Possible answers:
35%, 21, 60;
25% 09, 36;
34%, 17, 50

Students may wonder if the digits and resulting percentages should include decimals. Provide space for students to choose whether they think it should, as their responses will provide more entry points for learning.

5.9.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

Component Overview: *Students represent real-world ratio relationships with a table, double number line, and tape diagram. Have students complete each problem independently or with a partner, then choose representations to share with the class to provide students with an opportunity to check their work and discuss the different strategies.*

5.9.2 Guided Practice: Applying Ratio Relationships to Solve Percentage Problems

1. There is a 10% off sale on laptop computers. Ryan saved \$35 off of the original laptop price during the sale. Use the ratio table to determine the original price of the laptop.

Dollar Amount	Percentage
35	10
350	100

× 10 × 10

2. Lily was running in her first 10K race in Islamorada, Florida. She asked her sister to cheer for her when she was 75% of the way finished with the race so that it would motivate her to finish strong. The race is approximately 6.2 miles long. Use the number line diagram to determine how many miles Lily will have run when she is 75% finished with the race.

4.65 *miles*

3. During the first part of his hike along the Myakka Hiking Trail in Florida State Park, Andre drank 1.5 liters of water. If this is 40% of the water he brought on the hike, use the tape diagrams to determine how much water he brought.

water (liters)	0.75	0.75	0.75	0.75	0.75
percentage	20	20	20	20	20

Andre brought 3.75 liters of water with him on the hike.

Provide additional scaffolding for students using Math Nation's On-Ramp to 6th Grade video on decimals and place values.

Activate prior knowledge and facilitate student connections from Lesson 8.

After students have completed each question, choose representations to share with the class and engage the class in a discussion. Consider asking questions to deepen conceptual understanding like:

- *How do each of the three representations accurately show each ratio?*
- *Explain which representation or model is the most efficient for you.*
- *How are the representations related?*
- *How can you use each of these representations to determine missing values or percentages?*

5.9.3 Exploration: Using Tables and Unit Rates With Percentages

Component Overview: *In this Exploration, students explore percentages using real-world scenarios. Students begin by working with a partner to describe work shown in a ratio table. Students then complete tables to find equivalent ratios and their corresponding percentages of the whole. Students then use the tables to complete statements to interpret the different scenarios. Allow students to work through this component on their own then debrief their work before moving on to 5.9.4 Your Turn!.*

5.9.3 Exploration: Using Tables and Unit Rates with Percentages

A restaurant has a sign by the front door that says, "Maximum occupancy: 75 people."

Damian completed the first two rows of a ratio table representing this situation. His work is shown.

Number of People	Percentage of Occupancy
75	100
1	$\frac{100}{75} = \frac{4}{3}$

$\times \frac{1}{75}$ $\times \frac{1}{75}$

1. Work with your partner to complete the statements to describe Damian's work.

First, Damian interpreted the restaurant sign to mean that 100% is equal to 75 people.

Next, he multiplied both values in the ratio by $\frac{1}{75}$ to find the unit rate of the

- ○ percentage occupancy per 1 person
- ○ number of people per 1% occupancy

in the restaurant. He multiplied this factor by both values to find an equivalent ratio.

He then rewrote the result, $\frac{100}{75}$, as an equivalent fraction, $\frac{4}{3}$, by dividing both the numerator and denominator by 25.

If students have not mastered their multiplication and division facts, let them use a calculator to determine the ratios, rates, and percentages. The calculator will allow students to focus on mastering the learning goals of the lesson.

5.9.3 Exploration: Using Tables and Unit Rates With Percentages (continued)

2. With your partner, complete the table to find additional equivalent ratios. Use arrows and multiplication to show how you determine the values in each row.

	Number of People	Percentage of Occupancy	
	75	100	
$\times \frac{1}{75}$	1	$\frac{4}{3}$	$\times \frac{1}{75}$
$\times 9$	9	12	$\times 9$
$\times \frac{17}{3}$	51	68	$\times \frac{17}{3}$
$\times \frac{1}{68}$	$\frac{3}{4}$	1	$\times \frac{1}{68}$
$\times 40$	30	40	$\times 40$

3. Complete the statements to interpret the ratios.

When there are 9 people in the restaurant, it is 12 % full.

The restaurant is at 68 % capacity when 51 people are inside.

The restaurant is 40% full when 30 people are there.

4. Christian wondered if it is possible for the restaurant to ever be at exactly 1% capacity. Discuss his thinking with your partner. Then, write a brief summary of your discussion.

   Answers will vary.

   According to the ratio table, when the capacity of the restaurant is at 1%, there are ¾ of a person present. Answers should always be viable, and it is not possible to have ¾ of a person. This means that the restaurant will never have a capacity of 1%.

During this Exploration, circulate the room and observe students' informal discussions with their partners.

Listen for students who are using the formal vocabulary presented in previous lessons, such as unit rate and ratio, and connecting that to percentages. This illustrates conceptual understanding. Consider allowing these students to explain their work during the debrief at the end and highlighting the connections they have made.

Students may struggle to understand why the restaurant can never be at 1% capacity, because the idea of ¾ of a person may not make sense. Students may understand there can't be 1% capacity but may struggle to verbalize why. To help students verbalize their responses, ask probing questions like:

- *How can you determine how many people will be in the restaurant at 1% capacity?*
- *How can you use the table to help you find this?*
- *Would a fraction as the number of people be a reasonable answer here? Explain what you might do if the answer (number of people) results in a fraction.*

5.9.3 Exploration: Using Tables and Unit Rates With Percentages (continued)

Last Sunday, 1,575 people visited an amusement park.

- 56% of the visitors were adults
- 16% of the visitors were teenagers
- 28% of the visitors were children under the age of 12

5. Work with your partner to use the equivalent ratio table below to determine the number of adults, teenagers, and children who visited the park last Sunday. Show your work.
   Answers will vary.

Number of Visitors	Percentage
1,575	100
15.75	1
882	56
252	16
441	28

$\times \frac{1}{100}$ $\times 56$ $\times 16$ $\times 28$

Notice and Wonder

Consider engaging the class in a "Notice and Wonder" routine before completing question 6 to solidify conceptual understanding. What patterns do you notice in this equivalent ratio table? What do you wonder about this equivalent ratio table?

6. Complete the statements.

Last Sunday, 882 adults, 252 teenagers, and 441 children under the age of 12 visited the amusement park.

5.9.4 Your Turn!

Component Overview: *Students work independently to practice using equivalent ratio tables to calculate percentages in real-world contexts.*

5.9.4 Your Turn!

1. At a hardware store, a tool set costs $80. Using a coupon, Denise is able to pay $12 less than the original price.

   a. Complete the table.

	Dollar Amount	Percentage	
$\times\frac{3}{20}$	$80	100	$\times\frac{3}{20}$
	$12	15	

   b. The coupon gives Denise 15% off of the original price.

2. A bathtub can hold 80 gallons of water. Isaiah fills it up to 65% capacity to take a bath.

   a. Complete the table to determine how many gallons of water Isaiah had in the bathtub.

	Number of Gallons	Percentage of Capacity	
$\times\frac{13}{20}$	80	100	$\times\frac{13}{20}$
	52	65	

   b. Isaiah filled the tub with _52_ gallons of water to be at 65% capacity.

The ratios in this table are more complex than they have been throughout the lesson, so students may struggle to determine what they need to multiply the top row by to find the second row. To help students work through this component, ask clarifying questions like:

- *Where can you fill in the information you already know on the table?*
- *How can you determine what to multiply 80 by to find 12? Will this be a whole number, or a fraction? How do you know?*
- *What percentage does 80 gallons represent? Why?*
- *What can you multiply 100 by to find 65?*

5.9.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students demonstrate their ability to use equivalent ratio tables to calculate percentages in real-world contexts.*

5.9.5 Wrap-Up: Ticket Out the Door

1. A large school bus can hold a maximum of 72 children. This morning, Malachi's school bus only had 54 students on board. What percentage of Malachi's school bus is full?

Number of Students	Percentage of Capacity
72	100
54	75

$\times \frac{3}{4}$ $\times \frac{3}{4}$

Malachi's school bus was 75% full this morning.

Consider having students choose which representation they want to use to determine the percentage in substitution of completing the ratio table.

5.10 Converting Units Using Ratios

Lesson Introduction

Benchmark	MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.	Learning Target	• I can use ratios to convert units within the same measurement system.
Benchmark Coherence	**Building On** MA.5.M.1.1 Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.	**Working Toward** MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.	
Essential Questions	• How do you use ratios to convert units within the same measurement system? • How can you represent real-world scenarios using ratio conversions?		
Lesson Narrative	In this lesson, students use unit rates to convert various measurements of customary and metric units within those systems. Students explore unit rate conversions for length, volume, mass, and time and apply unit rates to convert measurements. Students analyze errors in another student's work converting units of measure. Then, students practice converting units of measure using rates. Finally, students reflect on their level of confidence using ratios to convert units within the same measurement system.		

Component Summary

5.10.1 Warm-Up (~5 min.)	Describe how furniture designers use math *(SE p. 219)*	5.10.4 Collaboration (10 min.)	Analyze errors in a student's work converting units of measure *(SE p. 222)*
5.10.2 Guided Instruction (20 min.)	Use unit rate conversions for length, volume, mass, and time *(SE p. 220)*	5.10.5 Your Turn! (5 min.)	Practice converting units of measure using unit rates *(SE p. 223)*
5.10.3 Guided Practice (5 min.)	Convert units of measure using unit rates *(SE p. 222)*	5.10.6 Wrap-Up (~5 min.)	Students reflect on their level of confidence using ratios to convert units within the same measurement system *(SE p. 224)*

Resources

Glossary	Materials	Additional Preparation
Key Terms: • Customary units • Metric units Additional Terms: N/A	• (Optional) Note cards • (Optional) Chart paper • (Optional) Markers	• Warm-Up 5.10.1 contains a video to accompany the question prompts.

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may make computational errors when converting units of measurement. • Students may accidentally use the reciprocal of the conversion factor.	• Students do not need to memorize conversion rates when engaging in problem-solving. Encourage students to use the tools they have (conversion tables provided as reference charts and in the Student Workbook) to solve problems efficiently.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Consider structuring 5.10.3 Guided Practice and 5.10.5 Your Turn! with the Think Pair Share routine. First, have students solve the problem individually. Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning. Discussing the problems will help students synthesize their understanding and build student confidence.	**MA.K12.MTR.2.1: Multiple Representations** In 5.10.2 Guided Instruction students use double number lines and equivalent ratio tables to convert units of measurement. Using these visual models will help students organize the quantities, determine the ratios, and accurately complete their calculations. Seeing multiple representations of the equivalent ratios will help students visualize the concepts and consolidate student understanding.
Differentiate Instruction	Consider alternative ways for students to engage with written descriptions or self-reflection like in 5.10.4 Collaboration and 5.10.6 Wrap-Up. Digital tools and apps like Flipgrid offer students differentiated approaches to communicating with others and capturing their personal reflections. 5.10.4 Collaboration: Have students verbally record their responses to Valerie using Flipgrid and submit as homework. 5.10.6 Wrap-Up: Have students describe what they learned with video, with more creative submissions earning bonus points. (encourage creativity in both examples)	

Lesson Notes:

5.10.1 Warm-Up: Furniture Designer

Component Overview: *Display the "Career Profile – Furniture Designer" video for the entire class. After students watch the video, prompt the students to read and answer the questions in the Warm-Up independently. Discuss how the furniture designer used math in the video with the class.*

5.10.1 Warm-Up: Furniture Designer

1. Jared Rusten is a woodworker who creates modern furniture in San Francisco.

   a. Jared usually starts designing furniture by sketching his ideas before he makes a scaled drawing using measurements to build the actual piece of furniture. What do you think a scale drawing is?
   Answers will vary.

   A scaled drawing shows a real object that has accurate sizes that are reduced or enlarged by a scale or specific amount.

   b. Jared said, "I never thought that I'd be using math as much as I do now." What are some of the ways he uses math as a furniture designer and maker?
   Answers will vary.

   Scale drawings allow us to visualize larger objects on paper. As a furniture designer and maker, you draft or sketch your designs on paper before creating the pieces. You are also able to scale your designs before they are in actual spaces. He multiplies to enlarge, divides to reduce, and uses area to ensure that the furniture fits in a space.

Turn and Talk
Engage students in a Turn and Talk, after watching the video, to discuss question 1. Position students to consider the different ways in which Jared used math to create furniture. This will increase investment and engagement with the lesson.

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Component Overview: *Students name different customary units of measurement for length, weight, volume, and time and find their unit conversions. Guide students through this component to explore how to convert units of measurement using the methods learned throughout this unit such as double number lines and tables.*

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships

Many different units are used to measure lengths, weight, capacity (volume), and time.

1. What are some units of measure that are used to take each type of measurement?
   Answers will vary, but could include:

Length	Weight	Capacity	Time
Centimeter	*Ounce*	*Liter*	*Second*
Foot	*Pound*	cm^3	*Minute*
Mile	*Kilogram*	*Quart*	*Hour*
Inch	*Gram*	*Pint*	*Day*
Meter	*Ton*	*Gallon*	*Week*
Yard		*Cup*	*Year*

The United States uses the customary system of measurement. The table shows the unit conversions for units within this system.

2. Label each row using the words **Length, Weight,** and **Capacity** to indicate what each set of units measures.

Type of Measurement	Unit Conversions in the Customary System
Length	$1\ foot\ (ft) = 12\ inches\ (in.)$ $1\ yard\ (yd) = 3\ feet\ (ft)$ $1\ mile\ (mi) = 5{,}280\ feet\ (ft)$ $1\ mile\ (mi) = 1{,}760\ yards\ (yd)$
Capacity	$1\ cup\ (c) = 8\ fluid\ ounces\ (fl\ oz)$ $1\ pint\ (pt) = 2\ cups\ (c)$ $1\ quart\ (qt) = 2\ pints\ (pt)$ $1\ gallon\ (gal) = 4\ quarts\ (qt)$
Weight	$1\ pound\ (lb) = 16\ ounces\ (oz)$ $1\ ton = 2{,}000\ pounds\ (lb)$

Consider activating student interest by prompting one of the following:

- *What types of units of measure do we think Jared (from the 5.10.1 Warm-Up) might use on a daily basis?*
- *Brainstorm some units of measure that your family members might use on a daily basis in their jobs.*
- *What is your dream job? What units of measure are used most often in that environment?*

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships (continued)

Each conversion rate represents a ratio.
Double number line diagrams can be used to convert units.

3. Use the double number line diagram shown to find how many feet (ft) are in 4 yards (yd).

There are <u>12</u> feet in 4 yards.

Tables can also be used to convert units of measure.

4. The average weight of an African forest elephant is $6{,}000$ pounds. Use the unit conversion and the table to determine how many tons the average African forest elephant weighs.

Pounds	Tons
2,000	1
6,000	3

× 3 × 3

Notice that each conversion rate is a unit rate.
Unit rates can be used as factors to convert from one unit to another.

For example, the factor $\frac{4\ quarts}{1\ gallon}$ is equivalent to 1 because the numerator and denominator are equivalent values. Multiplying by factors equivalent to 1 will result in equivalent values.

Think-Pair-Share
Consider implementing a Think-Pair-Share before students use the representations to convert the units.
Pose the following questions to students to activate prior knowledge and allow students to verbalize their thinking, which will engage auditory learners.
How can you use the double number line to determine how many feet are in 4 yards?
How can you use the equivalent ratio table to determine how many tons the average African forest elephant weighs?

5.10.2 Guided Instruction: Converting Units of Measure Using Ratio Relationships (continued)

Kaluba determined how many quarts are in 2 gallons of milk using this strategy. His work is shown.

$$\frac{2\,gal}{1} \times \frac{4\,qt}{1\,gal} = 8\,qt$$

Kaluba explained, "In my calculation, the unit 'gallons' is in the numerator and the denominator. This means I can eliminate this unit because it becomes 1, similar to when equivalent values are in the numerator and denominator. The result of the conversion is then the number of quarts."

5. Complete the calculation to find how many fluid ounces (fl oz) are in 3 pints (pt).

$$\frac{3\,pt}{1} \times \frac{2\,cups}{1\,pt} \times \frac{8\,fl\,oz}{1\,cup} = 48\,fl\,oz$$

The units of measure used in the United States are ***customary units***.

The units of measure used in most of the world are ***metric units***. Like the decimal system, the metric system uses the base 10.

The tables below show additional unit conversion rates for other units of measure.

Type of Measurement	Unit Conversions in the Metric System
Length	1 meter (m) = 100 centimeters (cm) 1 meter (m) = 1000 millimeters (mm) 1 kilometer (km) = 1,000 meters (m)
Capacity	1 liter (L) = 1000 milliliters (mL)
Weight or Mass	1 gram (g) = 1000 milligrams (mg) 1 kilogram (kg) = 1,000 grams (g)

Time Conversions
1 minute (min) = 60 seconds (s) 1 hour (h) = 60 minutes (min) 1 day = 24 hours (h) 1 year = 365 days 1 year = 52 weeks 1 week = 7 days

Challenge students who may finish early to convert units using the metric system or units of time. Consider asking students to determine the number of minutes in a week or a year (like in the musical Rent – 525,600). This will provide additional practice with more difficult conversion rates.

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

Component Overview: *Students work with a partner to convert units of measurement using unit rates.*

5.10.3 Guided Practice: Converting Units of Measure Using Unit Rates

1. Use the conversion rates to find each value. Show your work.

   a. 2.62 km $=$ 2620 m

   b. 9 in $=$ $\frac{1}{4}$ yd

   c. 1 week $=$ 168 hours

2. Adrian weighed 7 pounds (lb) and 4 ounces (oz) when he was born. How many ounces did Adrian weigh when he was born?
   116 *oz*

3. Ibuprofen is a medicine sometimes used for pain relief, such as for a headache. For adults, it is not safe to take more than 3200 milligrams (mg) of ibuprofen each day. How many grams of ibuprofen is safe for adults to take in one day?
   3.2 *grams*

Think-Pair-Share for MA.K12.MTR.1.1 Practice
Consider structuring this component with the Think-Pair-Share routine.

- *First, have students solve the problem individually, emphasizing their process rather than their product.*
- *Then, have students discuss their answers. Tell students to justify their responses and critique each other's reasoning.*
- *Discussing the problems will help students synthesize their understanding and build student confidence.*

5.10.4 Collaboration: Error Analysis

Component Overview: *Students work with a partner or small group to analyze errors in a student's work converting units of measure.*

5.10.4 Collaboration: Error Analysis

1. Valerie makes sure she drinks 8 cups (c) of water each day. She wondered how many gallons of water this is, so she used conversion rates to find out. Her work is shown.

$$\frac{80\ c}{1} \times \frac{2}{1} \times \frac{1}{2} \times \frac{1}{4} = \frac{160}{8} = 20\ gal$$

Valerie knew her answer didn't make sense because there is no way she drinks 20 gallons of water each day. She is not sure where she went wrong.

a. Discuss Valerie's work with your partner. Then, work together to find the correct number of gallons she drinks each day.

Valerie's first mistake was inputting 80 cups instead of her daily intake of 8 cups. Also, Valerie's first conversion factor, $\frac{2}{1}$, should be $\frac{1}{2}$ because there are 2 cups in 1 pint.

The correct answer is:

$$\frac{8\ \cancel{c}}{1} \times \frac{1\ \cancel{pt}}{2\ \cancel{c}} \times \frac{1\ \cancel{qt}}{2\ \cancel{pt}} \times \frac{1\ gal}{4\ \cancel{qt}} = \frac{8\ gal}{16} = 0.5\ gal$$

If students have not mastered the multiplication and division facts, let them use a calculator to determine the ratios rates, and percentages. The calculator will allow students to focus on mastering the learning goals of the lesson.

Circulate as students work with their partners to make informal observations and listen as students engage in discourse. Students should be using math vocabulary to find and explain the errors in Valerie's work, such as conversion, unit rate, and ratio.

5.10.4 Collaboration: Error Analysis (continued)

b. Help Valerie understand and learn from her mistakes. Write a brief note to Valerie explaining what her mistakes were and how she can learn from them. Include strategies that may be helpful.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Hello Valerie. I see that you do know your conversions! The one mistake you made in this problem is in the first conversion factor, $\frac{2}{1}$, which should be $\frac{1}{2}$ because there are 2 c in 1 pt. By writing $\frac{2}{1}$, you will not be able to eliminate the cups unit. One strategy that may be helpful is to include the units when writing the conversion factors. This will help you see that you are eliminating the correct units and using the correct conversion factors.

Students may struggle to explain Valerie's mistakes in writing or struggle to explain a strategy to fix them. Ask probing questions to guide student thinking or convert these questions into sentence stems for students to guide their responses.

- *What did Valerie do wrong? How did you fix that?*
- *What if you complete the question on your own and compare your work to Valerie's? Find the differences.*
- *How can you use units to help convert measurements correctly?*

c. Read your partner's note to Valerie and offer suggestions on ways to improve their note.
Answers will vary.

5.10.5 Your Turn!

Component Overview: *Students work independently to practice converting units of measure using unit rates.*

5.10.5 Your Turn!

1. Use the conversion rates to find each value. Show your work.

   a. $11{,}200$ milliliters $=$ 11.2 liters

   b. 3.5 gallons $=$ 56 cups

   c. 2 days $=$ $2{,}880$ minutes

2. Malik plays football for Jones High School in Orlando, Florida. Last Friday, he ran 94 yards for a touchdown on a kickoff return. How many feet did Malik run on the kickoff return?

   $$\frac{94\ yd}{1} \times \frac{3\ ft}{1\ yd} = \frac{282\ ft}{1} = 282\ ft$$

3. About 727 pints of milk are distributed each day from the cafeteria at Fort King Middle School in Marion County, Florida. How many gallons of milk are distributed at Fort King Middle School each day?

   $$\frac{727\ pt}{1} \times \frac{1\ qt}{2\ pt} \times \frac{1\ gal}{4\ qt} = \frac{727\ gal}{8} = 90.875\ gal$$

To encourage mathematical reasoning as students complete this component, ask questions like:

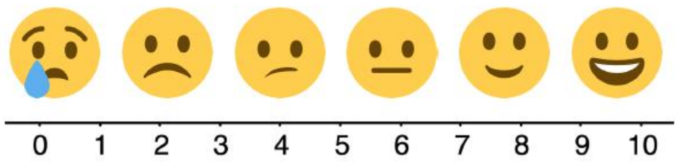
- *How can you determine the reasonableness of your answer using unit rates?*
- *How can you estimate what your answer should be before converting? Why would this be helpful?*
- *Will Malik have run more in feet or yards? How can you use this to assess the reasonableness of your response?*

5.10.6 Wrap-Up: Reflection

Component Overview: *Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson. This Wrap-Up assesses students' level of confidence with using ratios to convert units within the same measurement system.*

5.10.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with using ratios to convert units within the same measurement system.



Answers will vary.

2. Explain why you indicated the confidence level above.
Answers will vary.

Encourage students to get creative and represent their learning or progression from the lesson using video app platforms, drawings, or other methods and submit as homework.

5.11 Solving Problems Involving Ratios

Lesson Introduction

Benchmark	MA.6.AR.3.5 Solving mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.	Learning Target	• I can solve problems involving ratios, rates, and unit rates.

	Building On	Working Toward
Benchmark Coherence	MA.5.M.1.1 Solve multi-step real-world problems that involve converting measurement units to equivalent measurements within a single system of measurement.	MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions. MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Essential Questions	• How do you solve problems involving ratios, rates, and unit rates? • How do you convert units of measurement? • How do you calculate and compare rates? • Why is it important to find the unit rate when comparing different ratios?
Lesson Narrative	In this lesson, students work collaboratively to solve problems involving ratios, rates, and unit rates at three stations. At station 1, students begin by exploring whether various offers are a good deal based on the original and new price. Station 2 allows students to continue applying skills from the unit to convert measurements in a recipe from tablespoons to cups and vice-versa. At station 3, two texters are trying to determine who is the quickest; students must find and apply unit rates of each to declare a winner. This lesson culminates in a self-assessment to explain what students know and what they still have questions about.

Component Summary	5.11.1 Warm-Up (~5 min.)	Students determine if they can complete a task in a certain amount of time if they work at a given rate. *(SE p. 225)*	5.11.3 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. *(SE p. 227)*
	5.11.2 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. *(SE p. 225)*	5.11.4 Activity (10-12 min.)	Students work collaboratively in station activity to solve problems involving ratios, rates, and unit rates. *(SE p. 227)*
	5.11.5 Wrap-Up (~5 min.)	Students engage in meaningful self-reflection on growth mindset and learning progression from the lesson and the unit. *(SE p. 229)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms</u>: N/A	• Scientific Calculators • Cards for Station 1 • Devices for Station 3	• Station 1: Print and cut for blackline master activity; remove Task Card F

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may divide how many items there are by the price instead of dividing the price by how many items there are when finding the unit price.</td><td>The lesson has an Activity that consists of three different stations. Be sure to keep track of the time and ask students to rotate through the stations. Also, consider how students will check their answers. You could debrief all three stations at the end or have a key for students to self-check at each station.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Collect and Display
Students can engage in a Collect and Display for their stations in the Activity. Using chart paper to represent their strategies, approaches, and visual models provides students the opportunity to synthesize their learnings from the unit as they apply what they know in practice. Students can then engage in a Gallery Walk after to engage with the thinking of others.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students solve problems connected to real-world contexts in the Activity. Students see how calculating ratios, rates, and unit rates can help them determine the best value when shopping, convert ingredient measurements when cooking or baking, and compare rates. Each real-world context should spark student interest and motivate them to engage in solving the problems.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The Activity in the lesson consists of three stations at which students are applying concepts of ratios in a variety of real-world contexts. Consider a variety of strategies for engaging your students with Station Rotations.
Each small group completes each of the three stations. For small groups of 3-4 students per group, consider having 3 copies of each station for a total of 9 groups that could work simultaneously. After all groups have completed all three stations, have one student from each group pair up with a different group and share responses to verify accuracy and compare strategies.
Each small group is assigned one station at which they are experts, creating an anchor chart to display their work. After each group completes their work, every student then pairs with a student from another group and both students then act as experts or teachers of their station. Students repeat this process for the third station.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

5.11.1 Warm-Up: Bell Work

Component Overview: *Students determine if someone will be able to complete a task in a certain amount of time if they work at a given rate.*

5.11.1 Warm-Up: Bell Work

1. Mai has to fill and seal 60 packages of homemade trail mix to sell at the upcoming farmer's market. She filled and sealed 25 packages in 32 minutes.
   Mai wonders, "If I keep working at this rate, will I be able to get all of these done in less than 1 hour?"

   Explain whether Mai will be able to get all 60 packages filled in less than 1 hour working at this rate.
   The first step in determining if Mai would be able to fill and seal 60 packages in less than an hour, is to determine how much time it takes for one package to be filled and sealed.

   $$\frac{25 \text{ packages}}{32 \text{ minute}} = 0.78125 \textit{ packages per minute}$$

   The next step is to multiply the packages per minute by 60 to determine the amount of time it will take to fill and seal 60 packages.

   $$0.78125 \cdot 60 = 46.875 \approx 47 \textit{ packages}$$

   If Mai keeps working at this rate, she will not finish all 60 packages in less than an hour.

Consider allowing students to discuss their thinking before writing their answers to engage auditory learners. Encourage students to use a table or double number line to help them solve. This provides an entry point for visual learners.

5.11.2 Activity: Station 1: Is It a Deal?

Component Overview: *Students work collaboratively in the three following station activities to solve problems involving ratios, rates, and unit rates. The class can be broken into six small groups with two versions of each station. Students can either rotate through the stations or the stations can be passed from group-to-group.*

5.11.2 Activity: Station 1: Is It a Deal?

Your teacher will give you a set of cards showing different offers.

1. Find Card A and work with your partner to decide whether the offer on Card A is a good deal. Explain or show your reasoning in the space provided.

   Card A

   Original Cost per Water $= \frac{3.16}{4} = \$0.79$

   New Deal Cost per Water $= \frac{2.25}{3} = \$0.75$

   The original cost of the 4 pack of water is 4 cents more, so the new deal a better deal.

2. Next, split cards B–E so you and your partner each have two cards.
   a. Decide individually if your two cards are good deals. Explain your reasoning for your two cards in the table.

Is the new deal a good deal?			
Card B	**Card C**	**Card D**	**Card E**
Original $= \frac{3.50}{10} = \$0.35$ *New Deal* $= \frac{2.40}{6} = \$0.40$ *The new deal is not the better deal because the cost is lower in the original price.*	*Original* $= \frac{4.40}{5} = \$0.88$ *New Deal* $= \frac{3.12}{4} = \$0.78$ *The new deal is the better deal because the cost is lower than the original price.*	*Original* $= \frac{14.40}{16} = \$0.90$ *New Deal* $= \frac{9.00}{10} = \$0.90$ *Both deals have the same unit price, so neither is a better deal than the other.*	*Original* $= \frac{6.80}{8} = \$0.85$ *New Deal* $= \frac{5.22}{6} = \$0.87$ *The new deal is not the better deal because the original cost is lower than the new deal.*

All three stations provide an opportunity for multiple entry points and engagement of different learning styles.

Each station is set up to involve students in mathematical discussion with their partners or small groups, providing an entry point for auditory learners.

Each station also allows students to choose their own methods of representation. The visuals provided for Stations 1 and 3 offer an entry point for visual learners as well. Consider how to best engage kinesthetic learners as well, through the use of manipulatives or another hands-on strategy, like texting at Station 3.

Consider how to best facilitate student discussion at each station to ensure students are engaging with the material and deepening their understanding.

Circulate as students are working in their pairs or small groups and listen for discourse centered on using unit rates to determine the best deals, fastest texter, etc. If you notice students are not making this connection, ask probing questions to guide student thinking like:

- *How can you use what you know about ratios and unit rates to find the best deal?*
- *How will you know which deal is the best? Should the unit rate be larger or smaller than the original price?*
- *If you find the unit rate, how can you use it to find the cost of the item?*

5.11.2 Activity: Station 1: Is It a Deal? (continued)

b. For each of your cards, explain to your partner if you think it is a good deal and why.
The answers for cards B-E are in the above table.

c. Listen to your partner's explanations for their cards. If you disagree, explain your thinking. Write your partner's explanation for their cards in the table above, after resolving any differences.
Answers will vary. Sample responses in the above table.

d. Revise any decisions about your cards based on the feedback from your partner.

3. When you and your partner are in agreement about cards B–E, place all the cards you think are a good deal in one stack and all the cards you think are a bad deal in another stack.

4. Another potential deal is shown.

Fruit Snacks

Original: 12 for \$9.12
New Deal: 9 for ______

a. What is the price for 9 packs of fruit snacks at the same unit price as the original?

Original $= \frac{9.12}{12} = \$0.76$

$0.76 \cdot 9 = \$6.84$

b. Complete the statement.

Any price less than \$__6.50__ for 9 packs of fruit snacks would be considered a good deal because the cost per pack would be less than \$__0.72__.
Answers will vary. Any price less than or equal to \$6.84 will be a better deal.

If students have not mastered multiplication and division facts, let them use a calculator to solve the ratio, rate, and unit rate problems. The calculator will allow students to focus on mastering the learning goals of the lesson. Require students to write each calculation they use the calculator to find. This will reinforce their skills.

5.11.3 Activity: Station 2: Cooking with a Tablespoon

Component Overview: *Students work collaboratively in this station activity (#2 of 3).*

5.11.3 Activity: Station 2: Cooking with a Tablespoon

Diego is trying to follow a recipe, but he cannot find any measuring cups! He only has a tablespoon.

In the cookbook, it says that 1 cup equals 16 tablespoons.

1. How could Diego use the tablespoon to measure out these ingredients?

   a. $\frac{1}{2}$ cup almonds
   Note: 1 *tablespoon (Tbsp)*
   $\frac{1}{2}c \times \frac{16\ Tbsp}{1\ c} = 8\ Tbsp\ of\ almonds$

   b. $1\frac{1}{4}$ cup of oatmeal
   Note: $1\frac{1}{4} = \frac{5}{4}$

   $\frac{5}{4}c \times \frac{16\ Tbsp}{1\ c} = 20\ Tbsp\ of\ oatmeal$

   c. $2\frac{3}{4}$ cup of flour
   Note: $2\frac{3}{4} = \frac{11}{4}$

   $\frac{11}{4}c \times \frac{16\ Tbsp}{1\ c} = 44\ Tbsp\ of\ flour$

2. Diego also adds the following ingredients. How many cups of each did he use?
   a. 28 tablespoons of sugar
   $\frac{1\ c}{16\ Tbsp} \times \frac{28\ Tbsp}{1} = \frac{7}{4} = 1\frac{3}{4}c\ of\ sugar$

   b. 6 tablespoons of cocoa powder
   $\frac{1\ c}{16\ Tbsp} \times \frac{6\ Tbsp}{1} = \frac{3}{8}c\ of\ cocoa\ powder$

Think-Pair-Share
Consider structuring station 2 with the Think-Pair-Share routine. Consider including instructions or guidance for this routine either at the station itself if students are moving around the room, or on a white board that students can reference as they are working through each station.

This section requires several conversions using different ratios involving fractions. Some students may confuse how to write the ratio correctly, perhaps inverting the ratio, or they may write the mixed numbers as improper fractions incorrectly.
To address misconceptions, ask clarifying questions:

- *If you are converting cups to tablespoons, will the equivalent amount result in more or less tablespoons? How do you know?*
- *What ratio represents this? How can you write this ratio using a fraction bar?*
- *If you are converting tablespoons to cups, will the equivalent amount result in more or fewer cups? How do you know?*
- *What ratio represents tablespoons to cups? How can you write this ratio using a fraction bar?*

5.11.4 Activity: Station 3: Turbo Texting

Component Overview: *Students work collaboratively in this station activity (#3 of 3).*

5.11.4 Activity: Station 3: Turbo Texting

1. Jonah was with his friend Erin after school and he needed to text his mom. Erin watched Jonah text and commented on how slow he was.

   They decided to have a text race to determine who was faster. The image below shows what each person texted in the race.

   a. Discuss any initial observations with your partner.
   Answers will vary. Jonah's is longer than Erin's.

   b. Erin was able to send her text slightly faster than Jonah. Explain if you think this means she is the faster texter.
   Answers will vary. Erin's text did not have as many characters as Jonah's text, so she was able to send her text quicker. This does not make Erin the faster texter because she had fewer characters in her text.

   c. Describe the relationship you notice between the number of characters in a text versus how long it might take to type it.
   Answers will vary. The longer the test message, the longer is takes to send. Based on the number of characters listed, Jonah's text may take a little longer.

Advanced learners can be challenged to determine their texting rate and compare it to Jonah's, Erin's, or their classmates.

This component is centered on discussion and inquiry.

The discourse for question 1 should therefore sound more informal and inquisitive as students make predictions and assumptions before having concrete numbers to work with, leading them to determine what information they will need moving forward to conclude who is the faster texter.

d. What additional information would be helpful to determine who is the faster texter?
Answers will vary. To determine who is the faster texter, we would need to know how long it took both Jonah and Erin to send their texts.

2. The image below gives additional information about each person's text.

Jonah: $\frac{67}{7.4} \approx 9.05 \approx$ 9 characters per second

Erin: $\frac{46}{6.5} \approx 7.08 \approx$ 7 *characters per second*

Jonah is the faster texter because he sends approximately 2 *more characters per second.*

Work with your partner or small group to determine who is the faster texter, using your preferred strategy. Show your work.

Work shown above.

Some students may not realize they also need to know the amount of time it took each person to text to truly know who the faster texter is, assuming that whoever texted quickest must be the fastest. Ask guiding questions to ensure students arrive at this conclusion before completing the rest of the section:

- *Whose text is longer? How do you know?*
- *If Jonah and Erin send different length texts in different amounts of time, how can you determine who is the fastest?*
- *How can you apply what you know about unit rates to solve this problem?*

5.11.4 Activity: Station 3: Turbo Texting (continued)

3. Jonah and Erin decided to send one last text to see who could send it out faster. The image shows the text they each sent.

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

So let's settle this once and for all. I am a way faster texter and there is nothing you can do about it. I am the GOAT and you'll never catch me! I won!!

154 characters

If both people sent the message above at the same rate as before, predict about how many seconds it took each person to text the message to the nearest tenth of a second. Show your work.

Brother	Predicted Time
Jonah	17.0 *sec.*
Erin	21.8 *sec.*

Jonah:

$$\frac{154 \text{ } \cancel{characters}}{1} \times \frac{7.4 \text{ } sec}{67 \text{ } \cancel{characters}} = \frac{1{,}139.6}{67} \approx 17.01 \text{ } sec.$$

Erin:

$$\frac{154 \text{ } \cancel{characters}}{1} \times \frac{6.5 \text{ } sec}{46 \text{ } \cancel{characters}} = \frac{1{,}001}{46} \approx 21.76 \text{ } sec.$$

Encourage students to use any representation they have learned throughout this unit to solve. Student choice allows for multiple entry points and can also encourage students to make connections across representations as they work with their partners or groups to solve.

5.11.5 Wrap-Up: Reflection

Component Overview: *Students reflect on a question they have, a mistake they made, something they are unsure about, and something they are confident in from the lesson. Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson.*

5.11.5 Wrap-Up: Reflection

1. Complete the learning period based on today's lesson.

   Answers will vary.

One question I would like to have answered ...

One mistake I learned from today ...

One thing I am not sure about ... YET!

One thing I am confident I know is ...

What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of ratio and rate concepts addressed throughout the unit and how to best to address potential gaps in understanding prior to the unit assessment. Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.

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